

What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network

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Abstract

A relative-binding-free-energy network’s per-edge error bar, checked here against seed and velocity repeats on 1145 real binding edges, can serve as the null of a cycle-closure fit, turning a descriptive consistency check into a test with a stated null. What such a test can reach is fixed by the gradient-cycle decomposition of edge flows, due to HodgeRank: the cycle part of a network’s error is auditable and biases nothing, while the gradient part biases the affinities and is invisible to any statistic computed from the network’s own values, at any magnitude. The auditable subspace spans a third of the error space and, in the variance-weighted metric the closure statistic uses, carries under one per cent of the error against experiment, sitting a factor of 1.36 below chance on the worst system at the weakest reading tried. The test is correspondingly narrow. Its flags sit inside the standard fixed 1.0 kcal/mol hysteresis cutoff on every replicate; removing the edges it names does not recover accuracy against experiment; and widening every reported error by half, enough to cover the pose and preparation variance those repeats hold fixed, leaves nothing flagged. Resolving the invisible part requires measurements from outside the network, at worst one per unresolved dimension.

1 Introduction

Relative binding free energy calculations are the most accurate computational tool available for ranking how tightly a set of candidate molecules will bind a target, and structure-based drug discovery programs increasingly build them into decisions about which molecules to synthesize. That accuracy is expensive: each transformation between two related candidates, one edge of a perturbation network, requires hours of molecular dynamics simulation, and a realistic project touches dozens to hundreds of such edges. No program can run every edge to full convergence, so triage is unavoidable: some edges get simulated first, some get simulated longer, and some results get trusted enough to act on.

That triage is only as good as the error bar attached to each prediction. A result that looks converged but is not misleads a ranking of candidates; a result that is genuinely well converged but reported with an inflated error bar spends simulation time on a number that was already trustworthy. A surrogate built to speed up this triage inherits the requirement: every downstream use of the number, choosing what to simulate next, deciding when to stop, deciding which candidate to commit to synthesis, consumes the uncertainty as directly as the prediction.

The prevailing way to obtain that uncertainty is to learn it, the same way the prediction itself is learned: a head trained to output a mean and a variance [28], an ensemble of independently

trained models whose disagreement stands in for uncertainty [23], or a distribution-free wrapper calibrated after the fact on held-out data [2]. All three treat uncertainty as something to be inferred statistically from training examples. That approach overlooks a piece of the problem that is already solved: the estimator that turns raw simulation output into a binding free energy for a single edge, the Bennett Acceptance Ratio (BAR) estimator [7], is a textbook statistical estimator whose finite-sampling variance has a closed-form expression. That variance is a property of the estimator, not of the surrogate, and is computed from the same simulation output that produces the free-energy estimate, with no additional training data.

This work takes that closed form as the uncertainty a screening pipeline should report, and checks it against independent simulation replicates on a large set of real protein-ligand binding calculations. The calibrated bar then supplies a quality-control instrument. A perturbation network must close its thermodynamic cycles, and how far it fails to close them is already reported as a diagnostic; fitting that closure against per-edge precisions turns the diagnostic into a test with a stated null, one that flags networks whose simulations disagree with each other by more than sampling noise can explain.

The instrument raises a question that has not been asked of the diagnostic it replaces: what a cycle can see. The answer separates the part of a network’s error that an audit can reach from the part it cannot, bounds what an instrument of this kind can be asked for, and is measurable on public data. Both quantities are measured below: how much of a benchmark is auditable at all, and how much of the error real networks make against experiment falls where an audit could act on it.

This paper makes the following contributions.

- A reading of the gradient-cycle decomposition of edge flows from the error side. A perturbation network’s per-edge error splits into a part carried by the network’s cycles and a part that is a difference of per-ligand offsets. The second is unidentifiable from the network’s own edge values at any magnitude and is the part that biases the estimated affinities; the first is auditable and biases nothing. The per-edge share of audibility and the number of absolute measurements needed to resolve the rest follow from the same algebra. The decomposition is due to HodgeRank; the reading of it given here is what this work adds.
- A measurement of both quantities on a public replicate benchmark. The auditable subspace spans a third of the error space; on systems grounded against public affinity data it bounds the auditable share of the error against experiment well below chance. Two removal rules fail to improve accuracy against experiment, the second of them derived from the decomposition itself, and one of them does measurable harm on one system.
- A characterisation of the per-edge error bar a practitioner receives. The multistate Bennett acceptance ratio standard error, composed from the complex and the solvent leg, is checked against three independent protocol repeats on 1145 real binding edges. The two-state closed form behind it, the sandwich variance of the Bennett acceptance ratio estimator, is checked separately against bootstrap resampling of published work samples. Those samples all sit at high work-distribution overlap, so the check on real data does not reach the low-overlap regime, which rests on a controlled sweep instead.
- A quality-control instrument built on that bar. Fitting cycle closure with calibrated per-edge nulls separates the reproducible part of a network’s inconsistency from sampling noise. The per-edge residuals and the ranking of systems reproduce across independent repeats, and the list of flagged systems does not. A counterfactual bar as overconfident as a trained per-edge variance head destroys the separation, while the six per cent miscalibration a same-budget pooled estimator reaches leaves it intact, so the separation is scoped to heads of that severity.

- An applicability map across interval sharpness, edge ranking, synthesis-commit selection, reward-guided re-ranking and two decision tasks. None of the six puts the calibrated uncertainty ahead of a fair baseline. Commit trust on real out-of-distribution benchmark edges ties two standard post-hoc recalibrations of an overconfident uncertainty, and the stopping contrast in simulation separates only from a counterfactual comparison model, vanishing at the miscalibration a same-budget estimator reaches.
- The same closed form as the conductance of a measurement graph. The knowledge gradient is exactly zero along every direction of that graph that relative measurements leave undetermined, which follows from the graph Laplacian’s kernel and so holds for any positive conductances rather than from the sandwich weighting. Taken as a differentiable layer, the closed form has a backward pass costing a single division, and no experiment in this article trains through it.

The remainder of this article is organised as follows. Section 2 places this work against prior free-energy estimators, learned and recalibrated uncertainty, and acquisition methods. Section 3 gives the closed-form sandwich variance and its replicate validation. Section 4 presents the calibrated cycle-closure quality-control test on real binding data, the decomposition that bounds it, the measurement of where real error falls against that bound, and what survives a second replicate. Section 5 audits six downstream uses of the calibrated uncertainty and finds no gain over a fair baseline in any of them. Section 6 gives the data, the estimators, the models and the simulations behind every number reported above. Section 7 collects the limitations of the approach, and Section 8 concludes.

2 Related work

Free-energy estimators. BAR [7] and its multistate generalization MBAR [32] are the standard estimators for alchemical free energies; FEP+ [35] brought them to prospective potency prediction, and large retrospective assessments [30] and best-practice guides [27] document both their accuracy and their per-edge uncertainty, the sandwich variance of M -estimation [18, 36].

Learned and recalibrated uncertainty. The prevailing alternative is to learn predictive uncertainty: a mean-variance estimation (MVE) head trained by a Gaussian negative log-likelihood (NLL) [28], a deep ensemble for the epistemic term [23], the aleatoric/epistemic split [20], or distribution-free conformal prediction [2], including quantile-adaptive variants such as conformalized quantile regression (CQR) [29]. A predicted σ can also be *recalibrated* post hoc by temperature/Platt scaling [14] or isotonic regression [38].

Acquisition and reward re-ranking. Acquisition by the knowledge gradient [11] and its Monte-Carlo implementations [3, 12] drive edge selection, while GFlowNets [5, 26] sample candidates in proportion to reward; the resistance distance of a weighted graph Laplacian equals contrast variance in a Gaussian graphical model [21].

The works cited above use that uncertainty as a reported number or as a separately learned model. This work characterises the same sandwich variance against independent replicates, puts it to a distinct use as the calibrated null of a real-data cycle-closure quality-control test, and supplies it as a differentiable quantity that is also the conductance of a gauge-aware acquisition graph.

3 The per-edge error bar

Setup. For a single alchemical edge, forward and reverse works $\{W_i^f\}_{i=1}^{n_f}$, $\{W_j^r\}_{j=1}^{n_r}$ define unified coordinates $x_i = W_i^f$, $x_j = -W_j^r$. With $M = \ln(n_f/n_r)$ and $p(x; d) = s(x - d + M)$ (here $s(\cdot)$ is the logistic function, distinct from the uncertainty σ used later), the BAR estimate $\widehat{\Delta F}$ is the root of the score $S(d) = \sum_{j \in r} p(x_j) - \sum_{i \in f} (1 - p(x_i))$, and $\partial_d S = -\sum p(1 - p) =: -I(d)$, the curvature/overlap.

The layer’s backward pass costs one division by that curvature, by the implicit function theorem and without unrolling the root-find; the statement and its window-mean derivatives are Theorem S1 in the Supporting Information, where they sit because no experiment reported below trains through the layer. Everything below consumes the variance as a number or as a fixed graph weight.

Theorem 1 (Sampling variance via the sandwich). *$\widehat{\Delta F}$ is an M -estimator; its asymptotic sampling variance is the sandwich $\text{Var}(\widehat{\Delta F}) = A^{-1}BA^{-1} = B/I^2$, with $A = I$ and $B = n_f \text{Var}_f[p] + n_r \text{Var}_r[p]$ [18]. Here $A \neq B$ arises from BAR’s stratified two-sample design, not from model misspecification; the sandwich is the sample-computable form and agrees with the Bennett/Shirts–Chodera (MBAR) uncertainty [7, 32]. Concretely, the plug-in MBAR/Bennett uncertainty is $\text{Var}_{\text{MBAR}} = 1/I - (1/n_f + 1/n_r)$ (identical to `pymbar`’s MBAR uncertainty). Evaluated at the root, with works drawn from the correct pair of ensembles, this is not an approximation to the sandwich but the same quantity: $B/I^2 = 1/I - (1/n_f + 1/n_r)$ identically, so the $-(1/n_f + 1/n_r)$ term is precisely the $O(1/n)$ gap between the naive information-equality plug-in $1/I$ and the sandwich. The two sample plug-ins differ only through the moments they substitute, and that difference shrinks with n (paired Monte-Carlo check against the released implementation: the mean paired $|\text{Var}_{\text{MBAR}} - B/I^2|$ falls monotonically and by more than an order of magnitude from $n=40$ to $n=320$, the range verified here; beyond that the paired discrepancy is smaller than Monte-Carlo noise at a modest replicate budget, and the quoted range accordingly ends at 320). This is why bare $1/I$ (Figure 2, naive) overstates the calibrated uncertainty and `pymbar`’s reported MBAR se does not.*

Remark (scope of “calibrated”). B/I^2 is the variance of the *sampling* error of $\widehat{\Delta F}$, conditional on the works being drawn from the correct alchemical ensembles. It says nothing about systematic force-field bias, which is outside B/I^2 ; a separate epistemic correction term is attached in Section 5. Theorem 1 makes this term a differentiable closed form (Theorem S1) that is non-negative by construction, where the plug-in MBAR expression can return a negative value in a rare edge case (very high overlap with tiny samples, where the plug-in $1/I$ undershoots the $1/n_f + 1/n_r$ correction), whereas $B/I^2 \geq 0$ always.

The information equality $A = B$ (which would give $1/I$) holds for a single-sample MLE but fails for BAR’s two-sample design. The gradients are verified to machine precision and the variance to within 1% against 3000-replicate Monte-Carlo and `pymbar` in the good-overlap regime (at extreme low overlap, $n=20$, the bias-corrected plug-in runs $\sim 10\%$ below MBAR, both ≈ 1). The sandwich is derived for independent samples; on real FEP works both the sandwich inputs and the bootstrap truth are subsampled to the statistical inefficiency g [32] before B is formed, so the reported panel-B ratios use effective n .

The closed form against learned-variance baselines. Figure 2A sweeps the work-distribution overlap. The sandwich B/I^2 coincides with MBAR and Monte-Carlo truth across the range (reported/true se ≈ 1), with no Monte-Carlo. A fair learned mean–variance head, trained by Gaussian NLL on work-summary moments and on the overlap scalar I , is $\approx 7\times$ overconfident at a realistic budget of ~ 200 edges (reported se $\approx 0.15\times$ the truth, range $0.09\text{--}0.20\times$) and $\approx 5\times$ overconfident at

a twenty-fold larger oracle budget. A seed sweep and two corrected variants leave that unchanged: an objective fix does not recover calibration and an ensemble is unreliable in both directions at this budget (Supporting Information). The residual overconfidence is a property of this head’s per-edge parameterisation rather than of learning from labels: on the identical labels, pooling single-label edges by overlap recovers the sandwich to 6%, and a null miscalibrated by that much leaves the quality-control test intact, flagging seven systems of 48 against the shipped six. What the closed form provides over a learned per-edge σ is therefore not that the quantity cannot be learned, since the conditional $se(\text{overlap})$ is identifiable from single labels, but that the obvious per-edge parameterisation is badly conditioned where an analytic pooling is not. The bottleneck computes the variance exactly at no label cost, so the comparison is between a quantity learnable with data and one that is exact and free. Bare $1/I$ appears only as the information-equality plug-in it corrects ($1.08\text{--}2.65\times$, worst at high overlap), not as a baseline anyone reports. Figure 2B checks the same calibration on work samples from a real protein–ligand system, the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation (sandwich over decorrelated-bootstrap truth 1.00 , $[0.96, 1.07]$; naive $1/I$ over the same reference $2\text{--}12\times$, mean 6.07 , every window at overlap ≥ 0.84 ; per-window values, Supplementary Table S1). That reference reuses the same work samples, so the panel establishes agreement with a decorrelated bootstrap on real molecular dynamics in the high-overlap regime this schedule occupies, not a ligand-to-ligand binding quantity; the low-overlap behaviour is the controlled sweep of panel A and the per-transformation quantity is Figure 3. The replicate benchmark releases per-leg uncertainties and no work samples, so the work-sample route to B/I^2 is exercised on these `alchemtest` windows and not on the binding edges of Figure 3.

The reported bar against independent replicates. The validation that does not reuse its own samples, and the result the detector of Section 4 rests on, is Figure 3. Each edge’s reported bar is composed in quadrature from the complex-leg and solvent-leg MBAR uncertainties shipped with the benchmark, $se = \sqrt{se_{\text{complex}}^2 + se_{\text{solvent}}^2}$; each leg term is a multistate MBAR uncertainty over that leg’s λ windows, not the two-state BAR quantity of Theorem 1, so what is validated here is the bar as a practitioner receives it rather than the two-state identity itself. Across 1145 real binding edges from the OpenFE IndustryBenchmarks2024 [4] (3 independent replicates each), the reported MBAR se is calibrated-to-conservative against the across-replicate truth (reported/replicate-SD 1.41 , CI $[1.26, 1.62]$; conservative on 72% of edges and on 28 of the 34 systems carrying at least eight edges). Both sides of that ratio pool in quadrature, so the $n=3$ replicate term is unbiased at the variance level and needs no small-sample correction. The three replicates are independent stochastic repeats of the sampling process from identical starting coordinates (OpenFE `protocol_repeats` [4]); the across-replicate SD therefore reflects sampling/convergence noise alone and is a *lower bound* on full reproducibility (it excludes pose and system-preparation variance). The reported se does not systematically under-state real reproducibility, the opposite of the learned head’s $\approx 7\times$ overconfidence at realistic budget ($\approx 5\times$ even at large budget).

That conclusion disagrees with the closest measurement in the literature. Wade et al. [34] compare the analytic MBAR uncertainty against the spread over five independent replicas on 54 protein-ligand transformations across five targets, in two engines, and report that MBAR consistently under-estimates the error, by up to 0.9 kcal/mol. Two differences bear on the disagreement, in opposite directions. Their replicas resample the full ensemble, whereas the OpenFE repeats share starting coordinates, so the denominator used here is a lower bound on run-to-run variability and the ratio an upper bound on conservatism. Running their own functional on these edges sharpens the disagreement rather than resolving it: the analytic standard error falls below the replicate spread

on 0.279 [0.224, 0.332] of edges, against the 0.368 correctly scaled bars produce at three replicates, and the interval excludes it. Settling the question needs replicates that resample preparation and pose, which this benchmark does not provide.

Theorem 2 (Fisher–resistance correspondence). *For relative measurements $y_e = (\varphi_j - \varphi_i) + \varepsilon_e$, $\varepsilon_e \sim \mathcal{N}(0, V_e^{\text{true}})$ with $V_e^{\text{true}} = B_e/I_e^2$, the Gaussian posterior precision of the node potentials is the weighted Laplacian $L = \sum_e w_e (e_i - e_j)(e_i - e_j)^\top$ with conductances $w_e = 1/V_e^{\text{true}} = I_e^2/B_e$. Contrast variance equals effective resistance [21], a single edge recovers V_e^{true} , Sherman–Morrison gives $O(1)$ acquisition updates, and $\ker L \supseteq \text{span}\{\mathbf{1}\}$, so the knowledge gradient (KG) [11] is exactly zero on gauge/redundant directions: the acquisition is automatically gauge-invariant.*

The edge weights are the inverse sandwich variances, the readout of Section 3 entering the decision layer; the acquisition model identifies the reported variance with the true one ($c_e = 1$), an assumption the replicate validation of Section 3 tests rather than one this section adds. The gauge-zero property is structural: $\text{KG} = 0$ on gauge directions follows from the kernel inclusion alone, so it does not depend on the weight value, and by Gauss–Markov the value does not improve ranking (Section 5). For a *connected* network the inclusion is an equality, $\ker L = \text{span}\{\mathbf{1}\}$, for any positive conductances, and the overall additive offset is then the only unidentifiable direction; a network with k components has a k -dimensional kernel, one offset per component. The Laplacian-precision and resistance-distance machinery is classical generalized least squares (GLS) on graphs and the resistance distance of Klein and Randić [21]; what is specific here is the sandwich precision I_e^2/B_e as the conductance, in place of raw overlap, and the automatic gauge-invariance of the acquisition.

4 Calibrated cycle-closure quality control

This section separates the reproducible part of a perturbation network’s edge-level inconsistency from its sampling noise, using the bar of Section 3 as the null of a quality-control (QC) test. Candidate physical sources of a reproducible part include non-convergence, hysteresis, and charge, water or protonation artifacts, but this design does not identify which of them is present, and it does not separate them from a fixed consequence of the one starting structure every repeat begins from. The calibration underwriting the null is validated against seed and velocity replicate spread, a lower bound on full run-to-run reproducibility (it excludes pose and system-preparation variance), and the same restriction bounds what the detection claims below can mean: an inconsistency that survives repeats sharing their starting coordinates is established as more than sampling noise, not as physical error. This is *detection*, not weighting of a fixed estimate: the Gauss–Markov ceiling that makes the calibrated variance second-order elsewhere (in a connected graph the GLS contrast mean is unbiased for any positive weights, so the weight value cannot improve ranking, Section 5) does not bind a goodness-of-fit test, whose statistic depends on the weights directly.

A calibrated cycle-closure test. A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop is zero in expectation. Fitting node potentials by generalized least squares with conductances $w_e = 1/V_e^{\text{rep}}$, which Theorem 2 identifies with the sandwich precision I_e^2/B_e whenever the work samples are available (on this benchmark they are not released, so V_e^{rep} is the reported per-edge variance composed from the two legs, Section 6), gives residuals r_e and the goodness-of-fit statistic $X^2 = \sum_e r_e^2/V_e^{\text{rep}} \sim \chi^2(\text{dof})$, with dof the number of independent cycles (E minus the incidence rank). Under correctly-sampled physics $\mathbb{E}[X^2/\text{dof}] \approx 1$; here it runs below one (median 0.34), qualitatively consistent with the conservative bars of Figure 3. That median is not a conservatism factor, because the quantity the closure statistic identifies is not a median.

Two variances have to be told apart from here on: the variance a practitioner is handed, V_e^{rep} , which is what the fit uses as its weight, and the variance the edge actually has, V_e^{true} . They are related by $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$, so the per-edge calibration ratio c_e is the reported standard error over the true one: $c_e > 1$ is a reported bar that is too wide (conservative) and $c_e < 1$ one that is too tight (overconfident), and c_e is measured on this benchmark as the reported standard error over the across-replicate standard deviation. Whitening by the reported errors gives $\mathbb{E}[X^2] = \text{tr}(\Pi D)$ with $D = \text{diag}(c_e^{-2})$, so, as Theorem 3 below establishes, $\mathbb{E}[\chi_\nu^2]$ is the curl-leverage-weighted *mean* of c_e^{-2} . Inverting a median recovers a scalar conservatism only when c_e is constant across edges, and Figure 3B shows it is not. A ratio measured against seed and velocity repeats and a closure-implied one are therefore different functionals of a heavy-tailed per-edge ratio distribution, and no single scalar conservatism factor is identified by these data; the apparent disagreement between them is an artefact of comparing different functionals rather than a physical discrepancy. Residual correlation among edges sharing a ligand endpoint, which the diagonal- V GLS does not model, is separately negligible (excess correlation over the projector-induced null -0.0079 $[-0.0314, +0.0164]$). The Benjamini–Hochberg flag set is unchanged. A reduced χ^2 well above one signals edge-level inconsistency in excess of sampling noise. The test is (correctly) blind to node-consistent force-field bias, which cancels around a loop.

Theorem 3 (Estimation–detection conservation law). *In the whitened GLS fit let $\Pi = \mathbb{I} - H$ be the residual-maker projector onto the cycle space (Π , to keep M for the log sample-size ratio of Section 3 and R for a Pearson correlation). Its diagonal, the curl-leverage $h_e = \Pi_{ee}$, satisfies $h_e + w_e \Omega_e = 1$ with $w_e = 1/V_e^{\text{rep}}$ and Ω_e the effective resistance of edge e (Theorem 2); hence $\sum_e h_e = \text{dof}$, and the closure- χ^2 noncentrality against a systematic shift μ on edge e is $h_e \mu^2 / V_e^{\text{rep}}$. A bridge edge (in no cycle) has $h_e = 0$ and is undetectable at any μ ; a node-consistent (gradient) bias lies in $\text{range}(H)$ and is annihilated by Π . The QC’s blindness to node-consistent force-field bias is therefore a derived property, not an assumption, and h_e is a per-edge observability certificate with resolution $\delta_e^* = \sqrt{V_e^{\text{rep}}/h_e}$, the shift at which the closure statistic carries unit noncentrality.*

The algebra is standard: $h_e = \Pi_{ee}$ is the hat-matrix leverage of the whitened design [17], its complement $w_e \Omega_e$ is the correspondence between leverage and effective resistance that graph sparsification uses [33], with the resistance distance itself that of Klein and Randić [21], and $\sum_e h_e = \text{tr}(I - H) = \text{dof}$ is the trace identity for a projector. What the theorem adds is the reading of h_e as a per-edge observability certificate, which turns the test’s blindness to node-consistent bias from a modelling assumption into a consequence and names, before any simulation is run, the edges that can carry no evidence at all. The resulting map lists the structurally un-auditable (bridge) edges per system; on this benchmark the curl-leverage does not additionally predict the *location* of reproducible systematic error (studentized Spearman $\rho = -0.073$, permutation $p = 0.9571$, not significant; Supplementary Figure S5), so only the structural certificate is claimed.

Theorem 3 is the diagonal of a statement about the whole error space, and the whole statement is what bounds the method. Write the per-edge systematic error as a vector μ over the E edges. Whitening and projecting splits it, orthogonally and uniquely, into a part that lies in the cycle space and a part that is a difference of per-ligand offsets.

Theorem 4 (Auditable and unauditable error). *In the model $y = A\varphi + \mu + \varepsilon$ with A the signed incidence matrix of a network of N ligands, E edges and c components, and φ unknown: (i) for every δ , the parameter pairs (φ, μ) and $(\varphi + \delta, \mu - A\delta)$ induce the same distribution of y , so μ is identifiable only through $\Pi\tilde{\mu}$. The auditable subspace has dimension $\text{dof} = E - N + c$ and its orthogonal complement, the gradient fields $\mu_e = b_j - b_i$, has dimension $N - c$ and is unidentifiable: no statistic computed from a network’s own edge values distinguishes a per-ligand bias from no bias,*

at any magnitude. (ii) The bias such an error induces in the fitted potentials, $(\tilde{A}^\top \tilde{A})^+ \tilde{A}^\top \tilde{\mu}$, depends only on the gradient part, while the closure noncentrality $\|\Pi\tilde{\mu}\|^2$ depends only on the cycle part. The detectable part and the biasing part are exact orthogonal complements. (iii) For an error field with no preferential alignment, the expected visible fraction $\|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ is dof/E . (iv) Network estimation of $\Delta\Delta G$ under loop closure, with and without experimental constraints, is established [13, 37]; what the rank counting adds is the cost of the constraints. Given unbiased absolute measurements on a set S of ligands, the unidentifiable dimension is $(N - |S|) - c_{\text{free}}(S)$, with c_{free} the number of components holding none of them: the first anchor in a component removes only that component's arbitrary offset and no unidentifiable direction, each further anchor removes exactly one dimension, and resolving a per-ligand bias in full requires an absolute measurement on every ligand.

The decomposition is not new. Splitting an edge field on a graph into a gradient part and a divergence-free part, and reading the size of the second as whether the first is meaningful at all, is HodgeRank [19], where the gradient component is the global ranking being estimated and the cyclic component certifies it; a third, harmonic term appears there under a choice of two-cells and is not needed here. The hat-matrix and leverage machinery of part (ii) is likewise standard on networks of relative comparisons, where it already locates inconsistency in network meta-analysis [22]. That the closure check sees only cycles is in this sense a restatement, and it matches the long-standing reading of cycle closure as necessary and not sufficient [27].

The count in part (iii) is older still and closer to this setting than HodgeRank is. A network of relative comparisons under a consistency assumption has, in the language of mixed treatment comparisons, inconsistency degrees of freedom equal to the number of independent loops [25], which is dof here; the modern treatment of that model and its multi-arm complications is Higgins et al. [16]. Part (i), that a bias common to every comparison involving one node is absorbed into that node's effect and is invisible to any inconsistency test, is not stated in those terms in that literature, but the fact is immediate in its framework and may well be folklore there, as the corresponding fact is here, so the absence of an explicit statement is not evidence that the fact is new. The bias-adjustment strand of the same literature weakens the substance further. Bias-adjustment models in mixed treatment comparisons [9] treat study-level bias terms as quantities the network cannot identify on its own and inform them with external evidence or informative priors, which is part (i) for a different index and the same conclusion: a bias living where the design cannot separate it is resolved only from outside. The orthogonality of part (ii) and the anchor counting of part (iv) are likewise not located in that strand, which again is not evidence of their absence.

Part (ii) follows from part (i) in two lines. The bias in the fitted potentials is $(\tilde{A}^\top \tilde{A})^+ \tilde{A}^\top \tilde{\mu}$ and $\tilde{A}^\top \Pi = 0$, so the gradient part alone determines it while $\|\Pi\tilde{\mu}\|^2$ alone determines the noncentrality; it is the defining property of an orthogonal projector, read on the error rather than on the signal. What is left to this paper is that reading for alchemical networks and, above all, the measurement of where a real network's error falls between the two halves, which is Section 4.1 and what the rest of this paper rests on. Part (iv) is a dimension count and is not asserted as new. Network estimation of $\Delta\Delta G$ under loop closure is solved in Xu [37] by Fisher information and in Giese and York [13] by a constrained variational solution of the whole network; neither is the object part (iv) counts, which is the number of absolute measurements needed to remove the unidentifiable directions. That counting may well be present in either, and it is used here to state that cost.

Network maximum-likelihood estimation of $\Delta\Delta G$ networks [37], cycle-closure hysteresis as a diagnostic [8, 35], weighted cycle closure [24], and the constrained variational solution of a whole network under loop closure [13] are established; what is new is the *calibrated* null: a per-edge aleatoric variance measured against seed and velocity repeats rather than assumed, which gives the test per-edge adaptive thresholds and complements the fixed hysteresis cutoffs in standard use.

Those repeats share one pose and one system preparation, so the variance they validate excludes the component the alternative hypothesis is built from, and no controlled false-positive rate follows from them; nothing on this benchmark flags once the bars are widened to cover that component. The calibrated null comes together with the cross-replicate reproduction check of Figure 5B and the estimation–detection conservation law of Theorem 3, which fixes which edges are auditable at all; what is shared with the works above is the weighted least-squares treatment of the cycle graph itself. On the same 1143 edges a pre-registered head-to-head scored the calibrated per-edge null against the standard fixed 1.0 kcal/mol hysteresis cutoff and against a fixed-se χ^2 test, by area under the receiver-operating-characteristic curve; the criterion required beating both, and the verdict was a tie (Supplementary Figure S7). No further reading attaches to the ordering inside that tie. The only label these data supply divides each residual by the same per-edge variance the calibrated rule uses as its weight, so the ordering is monotone in proximity to that variance rather than in detection skill, and the comparison ranks nothing. The two rules are also not independent. On replicate 0 the three flag sets are strictly nested, the calibrated null flagging 6 systems, the fixed-se test 11 and the 1.0 kcal/mol cutoff 26, so the calibrated null catches nothing the cutoff misses and instead removes 20 of its flags. The chain does not survive the other repeats intact: on replicate 1 the calibrated null flags 6, the fixed-se test 13 and the cutoff 24, and two systems break the strict chain, **ciordia_retro** flagged by the calibrated null but not by the fixed-se test and **mup1** flagged by that test but not by the cutoff. On replicate 2 the chain holds again at 3, 11 and 24. The containment that matters here holds on all three: the calibrated set lies inside the cutoff’s. Whether that removal is an improvement is what the contaminated label cannot say, and adjudicating those 20 needs new sampling or new experiment, which public retrospective data cannot supply. The calibrated null’s advantage over a fixed cutoff is accordingly structural and not demonstrated here: per-edge adaptive thresholds, a null measured against seed and velocity repeats rather than assumed, and the observability map of Theorem 3. That measurement is not a controlled error rate, since the repeats hold pose and preparation fixed. The evidence that the flags are real remains the distribution-free cross-replicate reproduction check, which does not depend on the χ^2 scaling, and it needs only the per-edge *se* the benchmark ships: the detector’s novelty is the calibrated null, not the differentiable machinery. That null is conservative in the bulk, with a median reduced χ^2 of 0.34, so on a typical system the test under-flags. The median of the statistic does not bound the realized false-positive rate: Benjamini–Hochberg controls the false discovery rate through the behaviour of the null *p*-values near the rejection threshold, not through the median of the statistic, and the per-system calibration of Figure 3 is heterogeneous enough that a conservative bulk and an anti-conservative tail can coexist. The correction is applied across systems, and the guarantee is carried by Benjamini–Yekutieli [6], which assumes no dependence structure at all, so the independence Benjamini–Hochberg needs is not required. It costs a factor $H_{48} = 4.46$ in the threshold and returns the identical six systems on replicate 0 and the identical three on replicate 2, dropping one, **brd4**, on replicate 1 (Supporting Information). Positive regression dependence is a plausible remark rather than an assumption anything here rests on: the systems are fitted separately and share no edges, while sharing a force field and, within a target, a preparation. The level quoted is nominal rather than verified. The case for the flags therefore rests not on the χ^2 tail but on the distribution-free cross-replicate reproduction check below, which assumes no Gaussian null. A subtler concern is that four of the six flagged systems are themselves locally overconfident against their own across-replicate spread (Figure 3: **brd4** 0.76 $\times$, **faah** 0.90 $\times$, **bace** and **hif2a** 0.96 $\times$), so a flag could partly reflect that system’s own too-tight bars. Retesting each against the width its own replicates measure keeps all six at the per-system granularity and two, **bace** and **cdk8**, at the per-edge granularity this work pre-specified, which is the only self-calibrated arm holding its level (1/400, that is 0.003 with an exact binomial interval [0.000, 0.014], against a nominal 0.05) and so

the conservative answer (Supplementary Table S6).

The detector on the replicate benchmark. On the OpenFE IndustryBenchmarks2024 [4] (1145 edges over 49 systems, of which the 48 with ≥ 1 independent cycle, holding 1143 edges, admit the test; 3 replicates), the median reduced χ^2 is 0.34 and a Benjamini–Hochberg false-discovery-rate (FDR) test at 0.05 applied to the first replicate flags six chemically-sensible systems (**brd4**, **bace**, **faah**, **cdk8**, **hif2a**, **p38**; Figure 4A; **brd4**’s data set is a buried-water set, **bace** is protonation-sensitive). That particular set of six is a single-replicate draw and should not be read as a census; Section 4.2 measures how much of it survives the other two replicates.

One of the six has an identifiable cause. The benchmark labels a system by target, and eight of its names union edges from more than one separately released data set. Where those releases share ligands the union carries cycles present in neither release alone, and the cycle spaces sit orthogonally inside the union’s, so the closure statistic splits exactly. On **cdk8** the split is decisive: the union gives $\chi^2 = 77.7$ on 29 degrees of freedom, of which the two releases carry 31.1 on 23 between them, leaving 46.6 on the six cycles that span them, a reduced χ^2 of 7.8 at $p = 2.3 \times 10^{-8}$. Neither release is inconsistent on its own: **merck_cdk8** (51 edges, 20 degrees of freedom) gives $p = 0.218$, 0.865 and 0.189 on the three replicates, and **misc_cdk8** (12 edges, 3 degrees of freedom) gives 0.091, 0.992 and 0.985, the first being the smallest of the six and still well short of any threshold used here. The two share seven ligands, which is where the spanning cycles come from. **cdk8** is flagged because two independently prepared releases of the same target disagree where they meet, which is edge-level inconsistency beyond sampling and so is what the test is for, but it is not a statement about either release’s simulations. No other system carries across-release cycles that matter: in six of the eight the releases share no ligand and form disjoint components, and **tyk2**’s single such cycle is unremarkable ($p = 0.67$).

The grouping is therefore a choice the analysis makes, and the flag set is not invariant to it. Re-running the same replicate-0 test on the release’s own blocks gives 56 testable systems and again six flags, but a different six: **cdk8** leaves and **jacs_mcl1** enters. The grouping by target name was not pre-registered, and the Benjamini–Hochberg family size depends on it. The published list is the by-target one throughout. Supplementary Tables S4 and S5 give the per-replicate degrees of freedom, reduced χ^2 and adjusted q -value behind every flag. The detector’s value is entirely a function of calibration (Figure 4B): a uniform $\times 0.15$ shrink standing in for the learned head’s central overconfidence (Figure 2) flags 88% of systems (42 of 48), which leaves almost nothing unflagged and so no longer separates anything, the calibrated sandwich flags 12% (6 of 48), a $\times 1.3$ inflation one system of 48, and a $\times 2$ inflation none. These are flagged fractions and not false-positive rates: which of the flags are false is not known here (Section 7).

The count is not separable from the level, and the shipped operating point is the conservative one. The scale at which the reported bars are correct follows from the functional that governs the statistic rather than from a pooled ratio: the curl-leverage-weighted mean of c_e^{-2} is 0.853, so the bars are wide by $1.08\times$ and the calibrated scale is 0.92, the interval on that scale being $[0.79, 1.04]$, the square root of the $[0.63, 1.08]$ interval on $\mathbb{E}[\chi_\nu^2]$ of Section 4.3. Correcting every reported bar to a scale in that interval and re-running the test on the real data gives six flags at 1.04 and at 1.00, seven at 0.92 and nine at 0.79 (**make figSelf**). Two scales have to be kept apart here, and they are reciprocal. The calibration scale is $c = \text{se}^{\text{true}}/\text{se}^{\text{rep}}$, the quantity measured above as 0.92; a simulation that keeps the test on the reported bars must therefore draw its truth at c times them, which is a null $1/c$ times the truth. Quoted in c , the realized probability that the test flags anything when nothing is wrong is 0/400 at 0.79, 1/400 at 0.92, 18/400 at 1.00 and 40/400 at 1.04: as levels with exact binomial intervals, 0.000 $[0, 0.009]$, 0.003 $[0.000, 0.014]$, 0.045 $[0.027, 0.070]$ and

0.100 [0.072, 0.134]. Four hundred draws do not separate the two lower points from each other or from zero, so nothing is read from their ordering; what the grid supports is the upper end, where the interval clears the nominal level, and the direction between the ends. That simulation verifies the Benjamini–Hochberg implementation rather than the real-data operating point, which is why the level quoted above is nominal rather than verified; what it does establish is a direction, the level sitting below nominal at the measured calibration and above it at the other end. Because the bars are measured wide, the level across the calibration interval runs from essentially zero at 0.79 to 0.100 at 1.04, so the exposure that matters is not conservatism but the anti-conservative end, where a bar slightly tighter than the truth doubles the nominal rate. Sensitivity is exposed in the opposite direction: at $1.3\times$ one system of 48 flags and at $1.5\times$ and $2\times$ none do, so nothing on this benchmark survives a bar widened to cover pose and preparation variance. Widening changes the null being tested rather than moving the same test along it, and it is sensitivity, not validity, that is fragile on that side. That uniform shrink is a stress test rather than a realistic head, and the comparison is a counterfactual in a further sense: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check. The head’s own measured, overlap-dependent profile applied per edge flags 42 of 48, so the degradation is not an artifact of uniformity, and a shuffled control with a swept magnitude shows it is the magnitude that destroys the test while over-wide bars cost power the other way (Supporting Information). The demonstration is accordingly scoped to heads as overconfident as the measured one (reported se $5\text{--}11\times$ too small); it establishes that such heads saturate the test, not the shrink magnitude at which selectivity is first lost. The three independent replicates confirm the flagged inconsistency is reproducible rather than sampling: pooling the replicates shrinks the bars (se $\rightarrow$ se/ $\sqrt{3}$) yet increases the flagged systems’ reduced χ^2 toward the $3\times$ variance-components ceiling (Figure 4C), which occurs only for a component that repeats (single-replicate $\chi^2_\nu = 1 + \sigma_{\text{sys}}^2/V_{\text{samp}}$ vs pooled $1 + 3\sigma_{\text{sys}}^2/V_{\text{samp}}$, with σ_{sys}^2 the reproducible variance component and V_{samp} the sampling one; the ratio $\rightarrow 3$ iff reproducible-dominated). The argument fixes the component’s variance and not its origin: it is invariant to whether σ_{sys}^2 is physical error or a fixed consequence of the one starting structure the three repeats share, both being constant across repeats.

Cross-replicate reproduction of the flagged inconsistency. Greedily removing the flagged (highest- $|z|$) edges restores consistency (reduced $\chi^2 \leq 1$) in 3–7 edges, versus 12–37 removed at random (Figure 5A). That contrast illustrates the repair procedure and is not evidence, because it cannot fail: the statistic being driven down is $X^2 = \sum_e r_e^2/V_e^{\text{rep}}$ and the removal order is descending z_e^2 , which is that sum’s own per-edge summand, so a networked world with no localized systematic error whatever reproduces the same contrast. The evidential check is instead out-of-sample. Fitting each replicate’s network separately, the per-edge standardized residuals correlate across independent replicates ($r = +0.30$ to $+0.42$, $n = 1143$, each interval excluding zero under a bootstrap over systems; Figure 5B), where sampling noise alone would give zero, so what the flags identify is reproducible across runs rather than a property of one draw. The correlation carries the same restriction as the pooling argument: any per-edge quantity held fixed across the three repeats produces it, the shared starting structure is such a quantity, and what is established is therefore reproducibility and not a physical origin.

4.1 The reach of the audit on a network’s real error

Theorem 4 makes two quantities measurable that were previously matters of assumption: how much of a network is auditable at all, and how much of its real error the audit can reach. Both are answered on this benchmark in Figure 6.

Of the 1143 edge dimensions carried by the 48 systems with a cycle, 372 lie in the cycle space. Two-thirds of the error space is node-consistent and therefore invisible to every closure-based diagnostic, this one included. Forty-eight edges are bridges, carrying no evidence at any magnitude. Half the remaining edges cannot resolve a shift below 0.79 kcal/mol, and only 27% of all 1143 can resolve one of 0.5 (Figure 6A). Both figures overstate the instrument’s reach. δ^* is the shift at unit noncentrality and not a detection threshold: power there runs from 0.170 to 0.070 across the degrees of freedom present, and 80% at $\alpha = 0.05$ needs 2.8 to 4.7 times it, moving the median resolution to roughly 2.2–3.7 kcal/mol. It is also carried on the raw reported standard errors, where Section 4 puts the closure-relevant scale at 0.92 of them. Composing the two gives the operational numbers, which are worse than δ^* suggests. Only 6.6% of auditable edges reach 80% power against a 1.0 kcal/mol shift on that edge alone, at $\alpha = 0.05$ and before any multiplicity correction. At the level the test actually fires, a system’s most observable edge needs a median 1.35 kcal/mol for 80% power (quartiles 0.81 and 2.06) and its median edge needs 3.33; a third of systems could detect a 1.0 kcal/mol error somewhere, and 58% could detect 1.5. The regime this instrument operates in therefore begins at about the top of the band that relative-binding-free-energy decisions turn on, which is a sharper statement of the same limitation the rest of this section measures.

That is the capacity of the audit; its reach on real error is smaller, and measured on different graphs. Affinities cover part of each ligand set, so Figure 6B runs on each system’s grounded sub-network rather than on the full networks of Supplementary Table S3. As (N, E, c, dof) these are (24, 31, 2, 9), (41, 59, 1, 19), (34, 51, 1, 18) and (23, 26, 2, 5) for **cdk8**, **hif2a**, **p38** and **bace**, with auditable fractions 0.290, 0.322, 0.353 and 0.192 rather than the tabulated ones (**cdk8** is 0.46 on its full network); two are disconnected, which the fit absorbs into one offset per component.

All four were selected by the test, so what follows characterises the networks the test flags and not the benchmark, and one of the four, **cdk8**, is flagged by a cross-release join rather than by either release’s simulations. On those sub-networks the auditable share of the measured error against experiment is 0.6% pooled, against those chance levels (Figure 6B); per system 0.0065, 0.0017, 0.0086 and 0.0460. Dropping **cdk8** leaves the pooled figure at 0.6% (0.0060 against 0.0060), so the pooled figure does not rest on it; in the unweighted metric it moves from 5.45% to 6.10%. Leaving out one edge at a time moves them by 1.9, 1.7 and 2.6 times on the first three and by 17 on **bace**, whose range spans a factor of 23; the pooled figure is stable, the per-system ones are not equally so. Both sides are measured in the variance-weighted metric the closure statistic lives in, and the margin over chance depends on that choice, narrowing to a factor of two on **bace** under isotropy in kcal/mol and the pooled fraction rising to 5.45% in the unweighted metric, which recomputes the projector without variance weighting. Supplementary Table S2 gives all three conventions for all four systems with their own chance levels and their single-edge-deletion ranges; the weakest cell in it is **bace** isotropic under deletion, at 1.36 below chance. Against the frozen H1 criterion, a pooled degrees-of-freedom-adjusted gradient-model R^2 of at least 0.50 and positive on three of the four systems, the pooled value is 0.980 and the per-system values are 0.978, 0.995, 0.976 and 0.761, so the criterion is met. Pooling is over the four sub-networks jointly, the visible fraction as a ratio of summed squared norms and the adjustment carried on the summed $E = 167$ edges and rank 116, that is on 51 residual degrees of freedom, rather than as an average of per-system adjustments. The criterion evaluated is the degrees-of-freedom-adjusted rule the pre-registration froze rather than the weaker unadjusted form $1 - f_{\text{pooled}} \geq 0.50$. These support a bound and not a point estimate: the auditable share is small, at its weakest a factor of two below chance rather than the two orders of magnitude the pooled figure alone suggests. The measurement is not an artefact of the fit. It is a near-identity rather than a fit: per-ligand affinities make the experimental $\Delta\Delta G$ vector a gradient field, so $\Pi\epsilon$ reduces to the closure residual and the visible fraction is the closure statistic over the total squared standardized error. It also survives the obvious contaminants, holding on

medians rather than sums and holding on **bace**, whose ligands carry no enantiomer pair collapsed by the stereo-blind join (Supporting Information). Per-edge error against experiment runs 1.06–1.45 kcal/mol at the median against reported standard errors of 0.14–0.40: these networks are wrong by several standard errors while closing their cycles, in the one direction closure cannot look. Changing the label source to the curated per-edge affinities the benchmark of Section 5 carries, available for three of the four systems, agrees where it can be run (Supporting Information). The design does not license attribution. Experimental measurement error and the annotation join are per-ligand, hence gradient fields too, so what is established is that the dominant part of the error is node-consistent, not that it is force-field bias.

On **eg5**, whose ligands carry ChEMBL identifiers, the network $\Delta\Delta G$ (from the same GLS fit) agrees with real KIF11 experimental affinity [39] at a mean unsigned error (MUE) of 0.79 kcal/mol (95% bootstrap CI [0.59, 0.97], conditional on the fitted network; Pearson $R = 0.65$ [0.30, 0.83], $p \approx 2 \times 10^{-4}$, $n = 28$), and its clean cycle-closure reduced $\chi^2 = 0.64$ is consistent. That is not evidence that a clean system is accurate, and the five grounded systems cannot settle the question in either direction. The ordering by mean unsigned error is **eg5** 0.79 = **p38** 0.79 < **bace** 0.88 < **cdk8** 0.91 < **hif2a** 1.47, so the one unflagged system ties the best flagged one and is below the other three. That direction favours the flag, but it is a single unflagged system against four and it is not at matched grounding: **eg5** comes from a single named ChEMBL assay while the four flagged systems come from a target-wide connectivity search under the coverage rule. No separation claim follows from it in either direction. Acting on the QC does not improve accuracy here (internal $|z|$ vs error-against-experiment: $r = +0.13$), exactly as the scope predicts: on a clean system the residual error against experiment is dominated by node-consistent force-field bias, which cycle closure cannot see. The complementary test, that repairing a flagged system improves its accuracy, needs experimental affinities for the flagged systems; these are not directly tabulated in the replicate set but are recoverable by per-paper ChEMBL structure-search. Of the six flagged systems, four clear a 60% public-ChEMBL coverage gate (**cdk8**, **hif2a**, **p38**, **bace**; connectivity-based structure matching, one assay per system, IC50 before Ki, never mixed); their baseline network MUE against this grounded experimental affinity is 0.79–1.47 kcal/mol, typical-magnitude FEP error, confirming the grounding itself is sound. On these four, guided removal of the flagged high- $|z|$ edges (the same repair order as Figure 5A, fixing a removal count $K = 1$ –3 per system) does not reduce MUE-vs-experiment more than an equal-size random removal (Supplementary Figure S6; per-system one-sided permutation $p = 0.905, 0.136, 0.880, 0.071$; combined Stouffer $p = 0.48$, sign test $p = 0.69$; the frozen criterion is conjunctive, requiring Stouffer $p < 0.05$ and a guided Δ MUE above the 95th percentile of its own random null in a majority of grounded systems, and neither conjunct is met): two systems improve slightly over their random-null mean and two worsen slightly, with no system individually significant. The remaining two flagged systems (**brd4**, **faah**) did not clear the coverage gate and are not included.

That null is what Theorem 4 predicts and what the paragraph above measures. A removal rule reads the visible part of the error; the visible part carries under one per cent of what is wrong with these networks, and by part (ii) it is the part that biases the fitted potentials least. A second pre-registered rule tests whether the ranking was simply the wrong one, and fails to find that it was; a second failure to reject is weaker evidence than a rejection would be. Part (ii) suggests ranking edges by influence rather than by residual. The statistic used is $|z_e| \sqrt{(1 - h_e)/h_e}$ with $z_e = r_e / \sqrt{V_e^{\text{rep}}}$ the raw standardized residual, which is the classical influence form scaled by $\sqrt{h_e}$ rather than that form itself; edges with h_e below 10^{-9} are treated as bridges and given influence zero. Writing it in the unscaled form changes the removal set on two of the four systems and is reported with the outcome below. At the same removal counts it performs worse than the rule it was meant to replace, against

the same conjunctive criterion: better on one system of four (per-system ΔMUE -0.055 , -0.099 , -0.612 , $+0.140$ kcal/mol at one-sided permutation $p = 0.937$, 0.819 , 1.000 , 0.023 ; Figure 6C), at a pre-registered sign test of $p = 0.94$ against the residual rule’s 0.69 . On four systems that test attains only $\{1.0000, 0.9375, 0.6875, 0.3125, 0.0625\}$, so neither value could reach its own level and the two do not form an ordering, one system flipping separating them. The pre-registration’s other combination, a one-sided Stouffer, returns no usable number here, because one system’s permutation p saturates at the resolution of its own grid and Stouffer maps that to an infinite z ; clipping to the grid’s range gives 0.96 , and the residual rule, which has no saturated input, stands at 0.48 .

Neither rule beats random removal, and on one system the influence rule is worse rather than merely no better: on **p38** its guided ΔMUE of -0.612 kcal/mol falls below every one of the 1000 random draws (minimum -0.416 , no ties), so the opposite tail is at most $1/1001$. The tail is post hoc, the pre-registration having asked whether the rule helps; it is one system of four, about 0.004 under a Bonferroni allowance; and the magnitude is convention-dependent, -0.612 under greedy removal that refits after each edge and -0.232 when the statistic is applied once to the original fit; the sign survives both, and survives the strict influence form, which changes the removal set on two of the four systems but not here. The two edges removed carry $h_e = 0.099$ and 0.388 , so this is not the degenerate branch where a near-bridge is divided by its own vanishing leverage. The other tail is reported alongside it: on **bace** the same rule improves the error against experiment by 0.140 kcal/mol at a one-sided $p = 0.023$, which under the conjunctive criterion is not a result the summary can report, but which is exactly as post hoc and as single-system as the harm. Neither clears the pre-registered bar, and both are reported here rather than only the one running against the rule. The rule fails overall for the reason measured above: the influence form divides a residual by a small curl-leverage to recover the error of a single anomalous edge, and a smooth per-ligand field is not concentrated on any edge, so the division returns amplified noise.

The consequence for practice is a change of use rather than an abandonment. A flag remains a system-level signal that a protocol has produced results inconsistent beyond sampling, and that signal reproduces across replicates. It is not a repair target: deleting the edges it names cannot recover accuracy against experiment, because accuracy against experiment is set by a component the audit is blind to by construction. Reaching that component needs information from outside the network, and it has to be information the network does not already contain: an absolute binding free energy calculation carries exactly the force-field component of the per-ligand systematic error part (i) is about, so it resolves that component not at all; the mapping, soft-core, dummy-atom, schedule and restraint contributions are protocol-specific and would not transfer. A measured anchor resolves its dimension only to its own precision. By part (iv) the requirement does not soften when the anchors are unbiased: one anchor per component fixes only an arbitrary offset, and every dimension after that costs one more measured ligand.

Summary. The calibrated bar supplies, at no label cost, the null that turns cycle closure into a test separating reproducible from sampling inconsistency, and a bar as overconfident as the learned σ measured here destroys it. Theorem 4 bounds what that test, or any other of its kind, can be asked for, which is why its flags are not a repair list.

4.2 What survives a second replicate

The benchmark ships three independent repeats, so the flag set can be recomputed rather than assumed. The set does not reproduce. The eight-system union below is descriptive and carries no controlled error rate: three independent tests each holding a nominal 0.05 put the chance of at least one false flag among them at $1 - 0.95^3 = 0.143$ were they independent, which they are not; the

figure is illustrative in a second way too, since Benjamini–Hochberg controls the false discovery rate rather than the probability of any false flag. The counts are also per system and not per target, the benchmark’s 49 systems carrying 41 protein labels, with **bace** contributing four and **hsp90** four; the six flagged systems are six different proteins, so that count at least is not inflated by system splitting. The same pipeline at the same FDR level returns **{bace, brd4, cdk8, faah, hif2a, p38}** on the first replicate, **{bace, brd4, ciordia_retro, hif2a, p38, renin}** on the second and **{bace, cdk8, renin}** on the third: eight systems are flagged at least once, one (**bace**) is flagged every time, and pairwise Jaccard overlap is 0.50, 0.29, 0.29 against 0.056 for size-matched random draws, averaged over the three pairs. That instability is not a threshold artefact and persists from $\alpha = 0.01$ to $\alpha = 0.20$ (Supplementary Figure S8). Pooling the three replicates flags seven systems, the union minus **faah**.

The instability is not explained by how cyclic the networks are. Cycle counts on this benchmark are low: nine of the 48 systems carry exactly one independent cycle, twenty-three carry three or fewer, and the quartiles are 2, 5 and 12. A reduced χ^2 on one or two degrees of freedom is an unstable statistic, and those systems do vary more across repeats. But they are not the systems that move in and out of the flag set: seven of the eight ever flagged are well connected, and restricting the comparison to the best-connected third of the benchmark leaves the pairwise overlap no better than it is over all 48. The instability sits where the test has power, which makes it a property of the measurement rather than of thin networks.

Two properties do reproduce. The ranking of systems by closure inconsistency reproduces across replicates (Spearman 0.649, 0.644, 0.723 over all 48 systems), and the dependence of selectivity on σ -calibration, which is the claim this section is actually making, is essentially unmoved: the $\times 0.15$ shrink flags 42, 41, 41 systems, the calibrated bar 6, 6, 3, a $\times 1.3$ inflation 1, 1, 1, and a $\times 2$ inflation none, on the three replicates respectively. A single replicate supports a ranking and a calibrated operating point rather than a list of names. Among the published six, **faah** is the weakest on every measure: its reduced χ^2 crosses one in both directions across replicates (3.71, 0.45, 1.47), its standardized residuals barely reproduce between replicates ($r = -0.085$), and it is the one member the pooled fit drops, so it is a flag a second replicate does not support rather than a member of the set on the footing of **bace**.

4.3 Out-of-sample behaviour

Two checks use one replicate to predict another and so cannot be circular. First, the per-edge miscalibration measured from replicates two and three predicts the closure statistic of the replicate they exclude: with $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$ as above, Theorem 3 makes $\mathbb{E}[\chi_\nu^2]$ the curl-leverage-weighted mean of c_e^{-2} , and that prediction correlates with the observed per-system χ_ν^2 at Spearman 0.648 ($n = 48$). Both sides divide by the same reported variance, so the reference is not zero: under a simulation with no systematic error anywhere the same statistic returns 0.059 with a 95% upper limit of 0.364, and 0.648 exceeds every one of 200 draws (Supplementary Figure S9A). The aggregate level agrees less well: the predicted $\mathbb{E}[\chi_\nu^2] = 0.85$ [0.63, 1.08] sits below the observed degree-of-freedom-weighted value of 1.32 [0.81, 1.80], and the two intervals overlap without the prediction covering the observation. The scope of that agreement is narrow. The prediction is built entirely from measured miscalibration, so a rank agreement of this size says that the closure statistic largely reads out how well each system’s bars are scaled. It is a statement about what the test responds to, not evidence that the flagged systems are in error. The flagged systems do sit above the line, which is what any reproducible offset does: it inflates a closure residual without inflating the spread between repeats. Being reproducible across repeats that share a starting structure is what places a system above the line, and that is weaker than being in error.

Second, edges selected on one replicate are more inconsistent on the two held out than equal-size random selections are (mean gain in held-out reduced χ^2 , Stouffer $p = 1.3 \times 10^{-3}$, 1.3×10^{-3} , 4.4×10^{-3} over the three rotations), while a topology-only control built from curl-leverage is indistinguishable from random ($p = 0.21$ – 0.91). Each candidate statistic was calibrated against a no-error null before use and the realized level of every one is reported; two fired far above nominal and are not used. Only the consistency-gain statistic is specific to the flagged systems: an $|z|$ -rank-matched control drawn from unflagged systems reproduces the leverage-normalised residual signal but not the consistency gain. The effect is an aggregate one, significant individually in two of the five testable systems.

5 An applicability map for the calibrated uncertainty downstream

The detector of Section 4 runs on a network’s own residuals. Whether the same free calibration also improves downstream machine learning is a separate question, and on the six tasks audited here the answer is no: four performance tasks show no gain or outright harm, and the two decision tasks either tie a recalibrated baseline or separate only from a counterfactual comparison model that no estimator here produces. Reporting that null keeps the instrument of Section 4 from being read as a general claim about calibrated uncertainty. Supplementary Table S9 collects every effect against its baseline. Three performance tasks, interval sharpness, commit-to-synthesis selection and reward re-ranking over a measured congeneric set, run with the commit-trust decision task on the public OpenFF protein–ligand benchmark [15] (FEP+ and Merck data [30, 35], experimental $\Delta\Delta G$ labels, both-endpoint scaffold-disjoint out-of-distribution splits); edge ranking in active learning and the stopping decision run in controlled simulations. The amortized σ under test is the predictive uncertainty of the amortized $\Delta\Delta G$ predictor, the trunk, on unseen $\Delta\Delta G$, $\sigma_{\text{total}} = \gamma\sqrt{\sigma_{\text{ens}}^2 + \sigma_{\text{ale}}^2}$: deep-ensemble epistemic spread and benchmark aleatoric floor, conformally scaled at miscoverage $\alpha = 0.1$. It is not Figure 2’s per-edge BAR sampling-se or its learned-variance baseline; a bare “ σ ” below means σ_{total} .

Commit trust against recalibrated baselines. Fixing the ensemble mean and varying only the uncertainty in a commit rule (commit iff the lower-confidence bound $\mu - \kappa\sigma \geq \tau$), an overconfident epistemic-only σ over-claims badly: its realized-versus-claimed correctness shortfall is $+0.182$ $[0.132, 0.225]$ larger than the decomposed σ ’s (Supplementary Figure S15B). That number quantifies miscalibration; it does not rank the ways of curing it. On the same scale, temperature scaling [14] of that overconfident σ leaves $+0.034$ $[-0.018, 0.081]$ and split-conformal recalibration -0.029 $[-0.077, 0.007]$, both straddling zero in opposite directions. Trustworthy decisions need proper calibration of some kind, and all three routes supply it. Neither does this task separate them on cost, since all three fit a scalar on the same held-out split; the closed form’s cost advantage is the one Section 3 measures, a per-edge variance computed rather than learned, and none of these six tasks tests it.

The stopping simulation gives the same result. A calibrated learner’s claimed confidence tracks actual top- k correctness ($|\Delta| = 0.022$) and stops at 30 oracle calls 82% correct, while a learner whose assumed posterior standard error is scaled by the $0.15\times$ measured in Figure 2 stops twice as early at only 60% correct (Supplementary Figure S16). That second arm is a counterfactual comparison model built from a learned head’s measured miscalibration rather than an estimator run here, the same object and the same number Section 4 uses against the closure test, so what it leaves open is where on the assumed-error scale a real estimator would sit. Sweeping that scale answers it: at the $0.94\times$ a same-budget estimator pooling single-label edges by overlap reaches, the learner stops at

28.4 calls at the same 0.82 correctness; correctness at the stop first moves at $0.60\times$, and the halved budget needs $0.20\times$ (Supporting Information). The contrast is scoped, exactly as in Section 4, to per-edge heads that miss by $5\text{--}11\times$.

Sharpness, ranking and selection. On real held-out residuals of that trunk at matched ~ 0.90 marginal coverage the decomposed interval is slightly wider than flat split-conformal (width ratio $0.92\times$), because epistemic spread is small against the large irreducible out-of-distribution error; its conditional calibration is comparable to a fair Mondrian-by-target conformal and the advantage is not significant (ours-split -0.044 $[-0.121, 0.005]$; Supplementary Figure S14). The supplement’s synthetic illustration, whose generative structure the decomposition knows, is up to $2.2\times$ sharper than split-conformal; that does not transfer. For edge ranking, sandwich against naive weights differ by $0.98\times$ $[0.93, 1.03]$ on calls-to-top- k : in a connected graph the GLS contrast mean is unbiased for any positive weights, so the weight value is second-order for ranking. So too for commit selection at matched volume, where the σ -aware ranking does not beat a raw- μ ranking (-0.006 $[-0.016, 0.007]$). A calibrated bar informs whether to act and not which action ranks first.

Reward re-ranking on a measured congeneric series. A trajectory-balance GFlowNet [5, 26] samples a held-out target’s measured congeneric series under three rewards. On **eg5**, the on-disk target with the largest σ spread and so the one where a calibrated reward genuinely re-ranks candidates, the calibrated lower bound lowers the true hit-rate (0.140 against raw 0.249 ; cal-raw -0.109 $[-0.150, -0.072]$; Supplementary Figure S13). The high- σ ligands are the chemically novel strong binders ($\text{corr}(\sigma_{\text{total}}, \text{affinity}) = +0.42$), so the penalty $-\kappa\sigma$ demotes the true winners. Resting on one evaluable target this is a hypothesis rather than a transferable diagnostic: where σ flags novelty rather than quality, an uncalibrated reward is safer.

Synthesis. None of the six tasks puts the calibrated σ ahead of a fair baseline. On commit trust the physics σ matches post-hoc recalibration rather than beating it, and the stopping separation is absent at the six per cent a same-budget pooled estimator produces. The audit establishes a precondition rather than an advantage: decisions of this kind need proper calibration of some kind, an overconfident σ corrupts them, and the calibration demonstrably helps on real data inside the network, as the null of Section 4’s test, rather than downstream of it. The graph weight contributes the same way, through structure rather than magnitude, making the gauge directions exactly unidentifiable while by Gauss–Markov its value does not improve ranking.

6 Computational details

Data. Every closure fit consumes the release’s combined per-replicate table directly, taking the per-leg complex and solvent ΔG and $d\Delta G$ columns and composing the edge value and its standard error from them; no network-level or cycle-closure correction is applied to an edge before it enters the fit. The controlled panel of Figure 2 uses the **alchemtest** BACE1 relative-binding and benzene sets, from which forward and reverse work samples are available per λ window. The replicate validation, the cycle-closure test and every experiment on real binding data use the OpenFE IndustryBenchmarks2024 release [4], which ships three independent protocol repeats per transformation: 1145 edges across 49 benchmark systems. A system here is one of the benchmark’s named edge networks and not one protein: **jak2_set1** and **jak2_set2**, and **hsp90_single_ring** and **hsp90_kung**, are separate systems built on the same protein, so the system count exceeds the number of distinct proteins. The 48 systems with at least one independent cycle, which between them hold 1143 of the

1145 edges, admit the closure test, and each counts as one hypothesis in the Benjamini–Hochberg correction; the single remaining system is acyclic and contributes the other two edges. The per-system calibration ratios of Figure 3B are pooled instead over the 34 systems that carry at least eight edges, eight being the smallest edge count at which a ratio pooled against an $n = 3$ replicate spread is stable enough to plot on its own row; the other fifteen systems enter the aggregate ratio and the per-edge panel but are not shown individually. Grounding against measured affinity uses public ChEMBL IC₅₀ values. No free-energy simulation was run for this work; all alchemical output is taken as published.

Per-edge variance. $\widehat{\Delta F}$ is obtained by solving Bennett’s estimating equation $S(\Delta F) = 0$ by scalar root-find, and its variance from the closed form B/I^2 of Theorem 1 evaluated at the root. The backward pass uses the implicit function theorem, $\partial \widehat{\Delta F} / \partial \theta = (1/I) \partial_\theta S$, so it costs one division and does not unroll the root-find; gradients are checked against finite differences to machine precision.

Cycle-closure test. Node potentials are fitted by generalized least squares on the signed incidence matrix of the perturbation network, whitened by the per-edge standard errors, so that the fit is a weighted least-squares problem with conductances $1/V_e^{\text{rep}}$. Writing N for that incidence matrix, to keep B for the sandwich meat of Theorem 1, the statistic is the reduced χ^2 of the residuals with $\text{dof} = E - \text{rank}(N)$; per-system tail probabilities use the exact regularized upper incomplete gamma function rather than a Wilson–Hilferty approximation, and systems are flagged by Benjamini–Hochberg control of the false discovery rate at $\alpha = 0.05$ across systems. The population for the closure test is the 48 systems that carry at least three replicate-0 edges and at least one independent cycle, which between them hold 1143 of the benchmark’s 1145 edges.

The two baseline detectors. The fixed hysteresis cutoff flags a system when the largest absolute closure sum over a fundamental cycle basis exceeds 1.0 kcal/mol. That basis is the cotree of a spanning tree of the perturbation network, so it holds one cycle per degree of freedom of the closure fit, and the aggregation over them is a maximum. The rule consumes no per-edge uncertainty and applies no multiplicity correction. The fixed-se test is the closure fit above with one standard error per system, the median of that system’s per-edge reported values, substituted for every edge; its per-system tail probabilities pass through the same Benjamini–Hochberg control at $\alpha = 0.05$. The three detectors accordingly differ only in what they do with the per-edge bar: the cutoff ignores it, the fixed-se test replaces it by one number per system, and the calibrated null uses all of them.

Simulation protocols and model architectures. Two protocols underwrite numbers quoted above: the pure-null simulation that supplies every realized false-flag level, and the reproducible-systematic magnitude used as the label in detection comparisons. The amortized predictor used in the acquisition, stopping and reward experiments and the learned-variance baseline of Figure 2 are separate and much smaller models. Both protocols, and the features, layer sizes, objectives and training budgets of both models, are specified in the Supporting Information.

Software. Python ≥ 3.11 with numpy 2.5.0, scipy 1.18.0, pymbar 4.0.3, torch 2.12.1, RDKit 2026.3.3 and matplotlib 3.11.0. The pipeline is deterministic: in the environment recorded here, re-running a figure’s command reproduces that figure’s rendered raster byte for byte, and this was checked on the figures that depend on ensemble training rather than assumed. Ensemble training runs full batch on CPU for a fixed number of epochs, each member is seeded from its own index,

and the ligand ordering reaching the featurizer and the sampler is sorted rather than read out of a set, so no unrecorded source of variation enters the training path.

7 Limitations

The sandwich is aleatoric-only and conditional on correctly sampled ensembles. Systematic force-field bias lies outside it, and the out-of-distribution behaviour audited here rests on an ensemble plus a scalar aleatoric floor: there is no learned force-field-correction head, so the three-way decomposition of the synthetic illustration is not reproduced on real data.

The binding-edge se is validated against three independent replicates (Figure 3). In aggregate it is conservative by $\sim 1.41\times$, but calibration is heterogeneous: of the 34 systems carrying at least eight edges, 28 are conservative and six overconfident, namely **t4_lysozyme** $0.99\times$, **hif2a** and **bace** $0.96\times$, **faah** $0.90\times$, **brd4** $0.76\times$, and the protonation variant **bace_p3_arg368_in** $0.41\times$, the only one of the six that is not marginal. Those replicates are seed and velocity repeats from identical starting coordinates, so the across-replicate SD is a lower bound on full reproducibility, excluding pose and preparation variance, and the measured conservatism may be closer to neutral against real reproducibility. The same restriction limits what the quality-control test detects: a pose or a protonation assignment carried unchanged into all three repeats reproduces exactly as physical error does, so the flags are reported as reproducible inconsistency throughout.

The test detects only *edge-level* inconsistency, and neither removal rule improves accuracy against experiment: guided removal ties random on the four grounded systems (Stouffer $p = 0.48$; Figure S6) and the influence rule does no better overall, doing measurable harm on **p38** and reaching a nominal improvement on **bace**, neither clearing the conjunctive criterion (Section 4.1; Figure 6C). Theorem 4 offers a structural reading of both nulls, bounded by the margin Section 4.1 measures rather than established independently of it. The pre-registered falsifier for that reading is a guided removal beating its random null in a majority of grounded systems at a combined p below 0.05, which would show the visible part actionable; it is not met. The measurement rests on four systems selected by the test, so it characterises those rather than the benchmark, and for **cdk8** not even that: its grounded sub-network is all 31 edges of **merck_cdk8**, the block whose own closure statistic is not significant on any replicate, so the evidence that flagged it lies outside the sub-network measured. **p38**’s grounded sub-network is the **jacs_p38** block, which is individually significant ($\chi^2_\nu = 2.63$, $p = 1.9 \times 10^{-4}$), and **hif2a** and **bace** are single-block systems. It runs on a connectivity-based ChEMBL join, stereo-blind, so enantiomer pairs sharing a 2D graph collapse to one experimental value, inflating absolute MUE while cancelling in the guided-versus-random contrast since both arms read the identical cached affinity; the experimental and annotation errors it carries are per-ligand quantities entering the same unauditable subspace as force-field bias, so the measurement locates the error without attributing it. The influence rule is a second pre-registered hypothesis on the data that returned the first null, so it is a second look and not independent confirmation. The theorem bounds what a network’s own edge values can support and says nothing against methods bringing information from outside the network. A prospective flag, repair and remeasure test with new molecular dynamics lies outside what public data can support.

The cross-replicate check shows that what is flagged reproduces across independent runs; it is not a false-positive-rate guarantee and does not certify the nominal Benjamini–Hochberg rate. Recall cannot be measured at all, there being no ground-truth error labels. Replacing each system’s reported standard error by its own replicate-measured calibration changes the flag set in both directions, by an amount depending on the granularity and the direction, from retaining two of the six to reaching sixteen. The arms, their realized false-flag probabilities and the single one that

holds its level are tabulated in the Supporting Information, and none is reported here as a flag set. The flag set is therefore sensitive to calibration error of the size actually present, in both directions, which limits how far any single flag list should be trusted.

Together with the median reduced χ^2 of 0.34, which implies limited single-replicate power, that makes the flagged systems a floor rather than a prevalence estimate. `renin` carries the reproducible signature by the pooled-replicate ratio, its reduced χ^2 rising to 4.7 pooled from a per-replicate mean of 2.5, and clears the false-discovery threshold on two of the three replicates though not on the first, which is the one the published list is drawn from; it is a system that list misses rather than one the test misses, and `ciordia_retro` is a second such case.

The gauge-acquisition and stopping experiments are controlled simulations, and the reward re-ranking audit rests on a single σ -varying target, so its conclusion that σ tracks novelty rather than quality is directional and should be tested across more chemotypes.

8 Conclusion

A perturbation network’s own edge values determine only the cyclic half of its error. Fitting cycle closure against the precisions the free-energy estimator already reports turns a descriptive consistency check into a test with a stated null, orders networks by how far their own calculations disagree, and names in advance the edges that lie in no cycle and can carry no evidence. The instrument consumes only quantities the estimator computes anyway, and its selectivity is set by how well those quantities are calibrated. On these data the test catches nothing the fixed hysteresis cutoff misses: the two flag sets are nested and the head-to-head between them is a tie, so the advantage is structural, being per-edge adaptive thresholds, a null measured against seed and velocity repeats, and the observability map. It is also not a repair list. Removing the edges it names does not recover accuracy against experiment, and a rule built from the theorem does no better and on one system does harm, because the component that sets accuracy against experiment lies in the half no closure statistic reaches.

Closing that gap needs information the network does not contain, and in absolute measurements the worst case costs one ligand per dimension. That is a worst case: a bias smooth in chemical space is a far smaller object, and a noisy anchor resolves only to its own precision. A second ceiling sits above all of this and is not a property of the decomposition. The bar the test uses as its null is measured against repeats that resample seeds and velocities while holding one pose and one system preparation fixed, so it prices sampling noise and not the preparation an entire network inherits. Widening it by half, which is what covering that would take, leaves nothing on this benchmark flagged at all. What the instrument can reach is bounded twice over: once by the geometry of the network, and once by a variance component its own null never sees. The practical use of cycle closure is accordingly as a calibrated alarm, with the error that moves a ranking to be sought in absolute measurements or independent preparations instead.

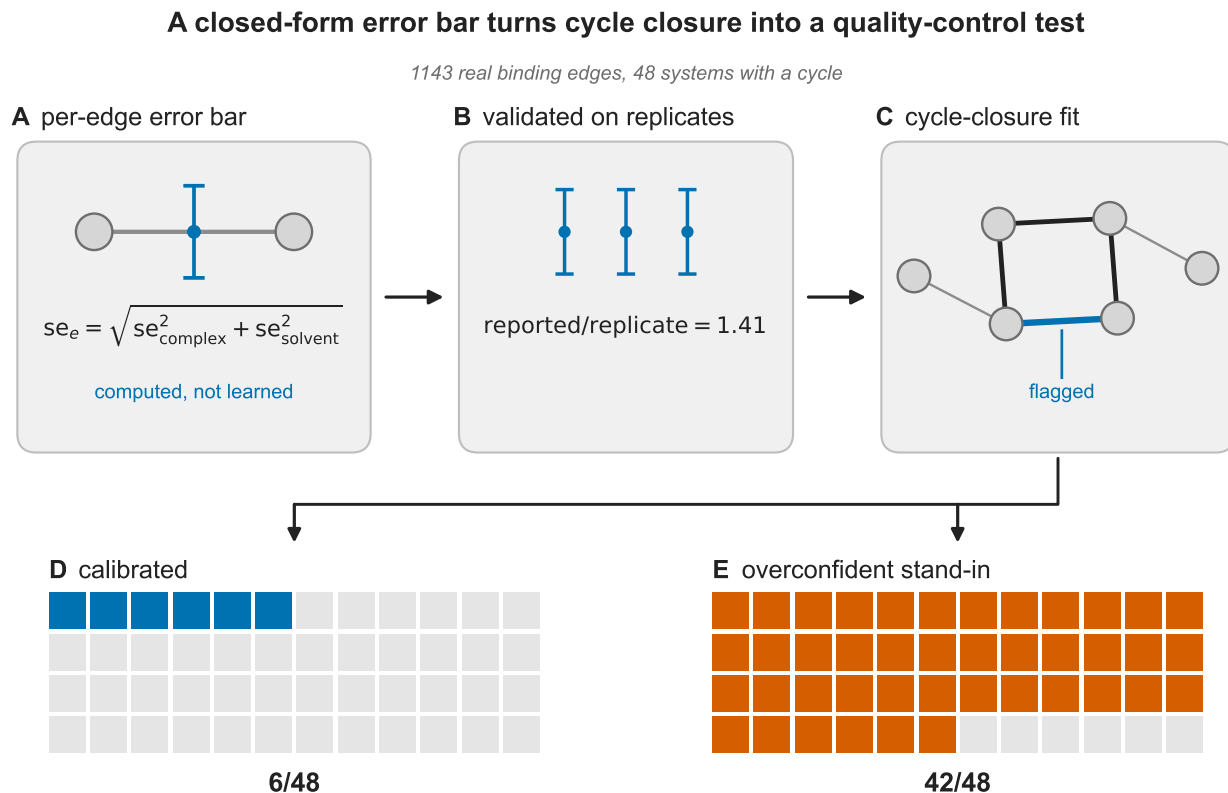


Figure 1: **Overview.** The pipeline this article runs on the benchmark, left to right. Every edge arrives with a per-edge standard error the estimator computes rather than learns, composed in quadrature from the complex-leg and solvent-leg uncertainties; that bar is validated against the benchmark’s three independent protocol repeats (reported over replicate spread 1.41); and it is then the null of a generalized-least-squares cycle-closure fit, drawn as a small perturbation network with one cycle highlighted and one edge marked as flagged. Across the 1143 real binding edges that lie in the 48 benchmark systems carrying a cycle (of 1145 edges over 49 systems), that calibrated bar flags 6, 6 and 3 systems on the three repeats, one of them on all three, and a comparison bar built edge by edge from a learned head’s measured miscalibration profile, not that head run here, flags 42.

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Author Contributions

N.L.P. is the sole author and is responsible for the conception and design of the study, the theory, the software, the analysis, the preparation of the figures, and the writing of the manuscript.

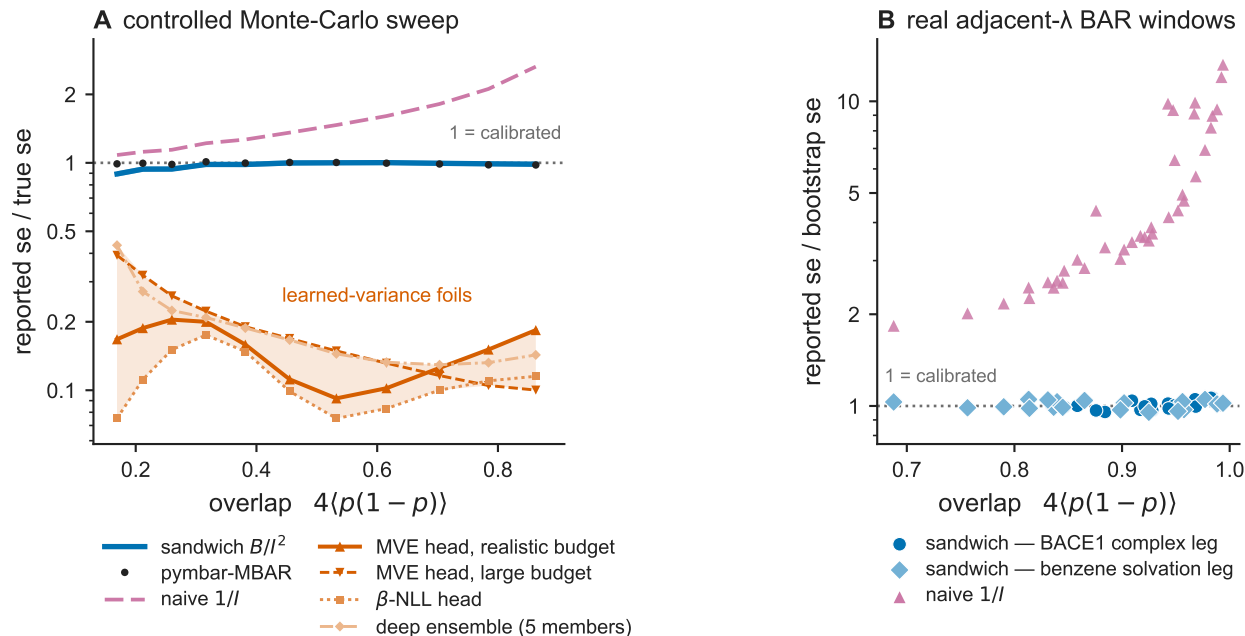


Figure 2: **Sandwich variance against learned-variance baselines, controlled and real data.** (A) Reported standard error divided by the true standard error, versus work-distribution overlap, for the sandwich B/I^2 , pymbar-MBAR, the naive information-equality plug-in $1/I$, a learned mean-variance (MVE) head at a realistic and at a large training budget, a β -NLL head, and a 5-member deep ensemble, in a controlled Monte-Carlo sweep. (B) Reported standard error versus decorrelated-bootstrap standard error on the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation (filled circles) and on the 19 windows of the benzene solvation legs (filled diamonds), for the sandwich B/I^2 and the naive $1/I$. Each point is one two-state BAR estimate between neighbouring λ values.

Conflicts of Interest

The author declares no competing financial interest.

Data and Software Availability

All data used in this work are public. The `alchemtest` BACE1 and benzene alchemical sets supply the work samples of Figure 2; the OpenFE IndustryBenchmarks2024 replicate set supplies the 1145 binding edges and their three protocol repeats; the OpenFF protein-ligand benchmark, which aggregates the FEP+ and Merck data sets, supplies the downstream audit tasks; and ChEMBL supplies the experimental affinities used for grounding. No free-energy simulation was run for this work; all alchemical output is taken as published. The BAR layer, the cycle-closure quality-control module, the theorem-verification and quality-control unit tests, and every figure-regeneration script are released as open source under a permissive licence, with each figure in this article regenerated by a single documented command. The code is available at <https://github.com/doctawho42/fep-surrogate-pipeline> under the MIT licence; the version from which every figure in this article was generated is tagged v1.0.0. The archived release is deposited at <https://doi.org/10.5281/zenodo.22046693>.

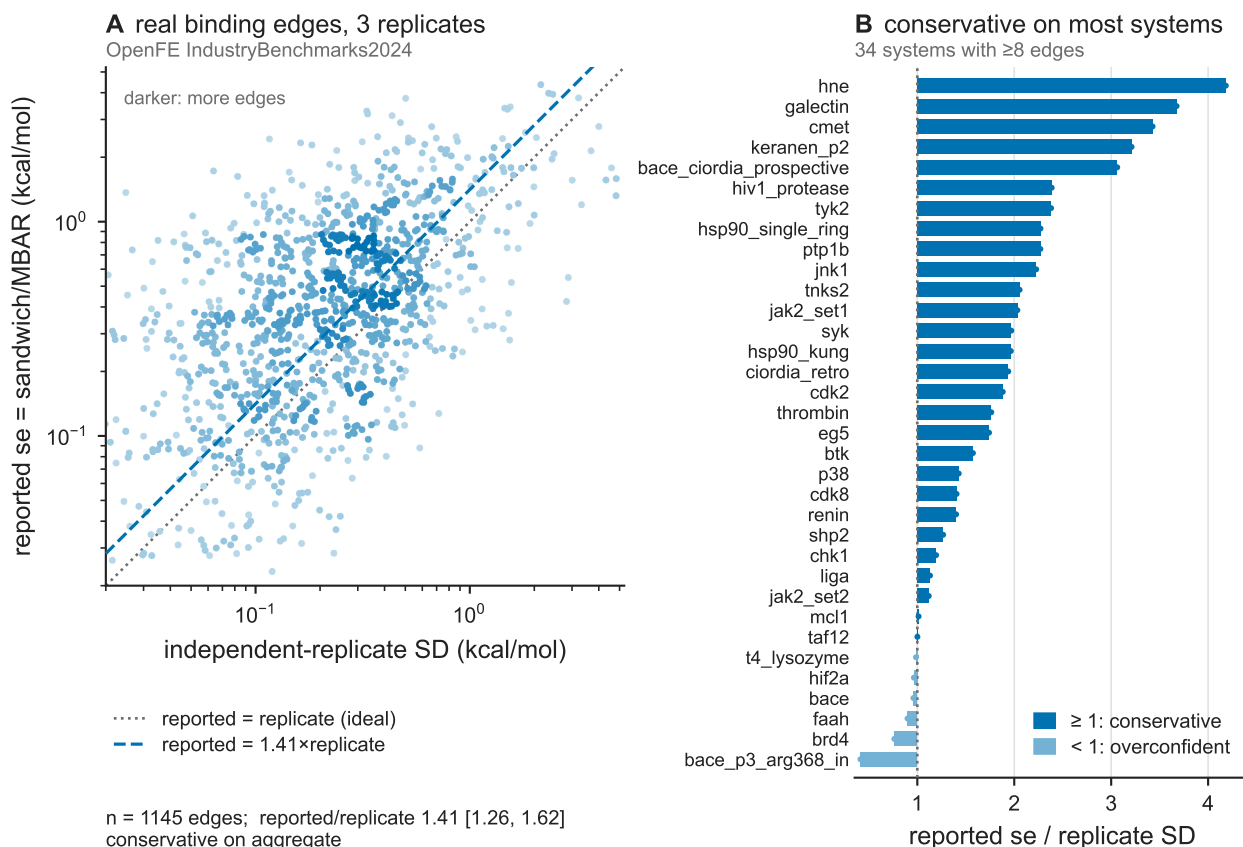


Figure 3: **Independent-replicate validation on real binding edges.** Reported per-edge standard error versus the across-replicate empirical standard deviation, on 1145 OpenFE Industry-Benchmarks2024 [4] edges (3 independent replicates each). (A) Reported standard error versus across-replicate standard deviation, with the $y=x$ line. (B) Per-system pooled ratio of reported standard error to across-replicate standard deviation, for the 34 benchmark systems carrying at least eight edges.

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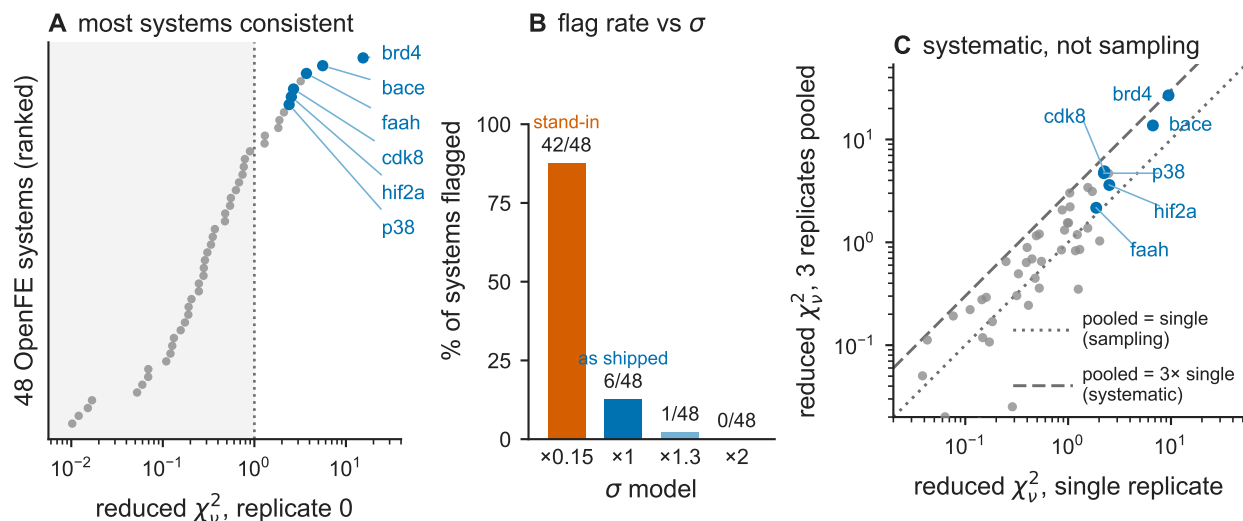


Figure 4: **Calibrated cycle-closure quality control.** (A) Per-system reduced χ^2 from a generalized-least-squares cycle-closure fit, ranked, with shading below $\chi^2_v = 1$ and the flagged systems marked. (B) Percentage of systems flagged under four σ models, in plotted order: a uniform $\times 0.15$ shrink, the reported bar as shipped, a $\times 1.3$ inflation and a $\times 2$ inflation. The shipped bar is the tick marked $\times 1$; the scale at which those bars are correct is measured as 0.92 of them, which is a different quantity from the tick. (C) Reduced χ^2 from a single replicate against the value from the three replicates pooled, with reference lines at pooled/single = 1 and = 3.

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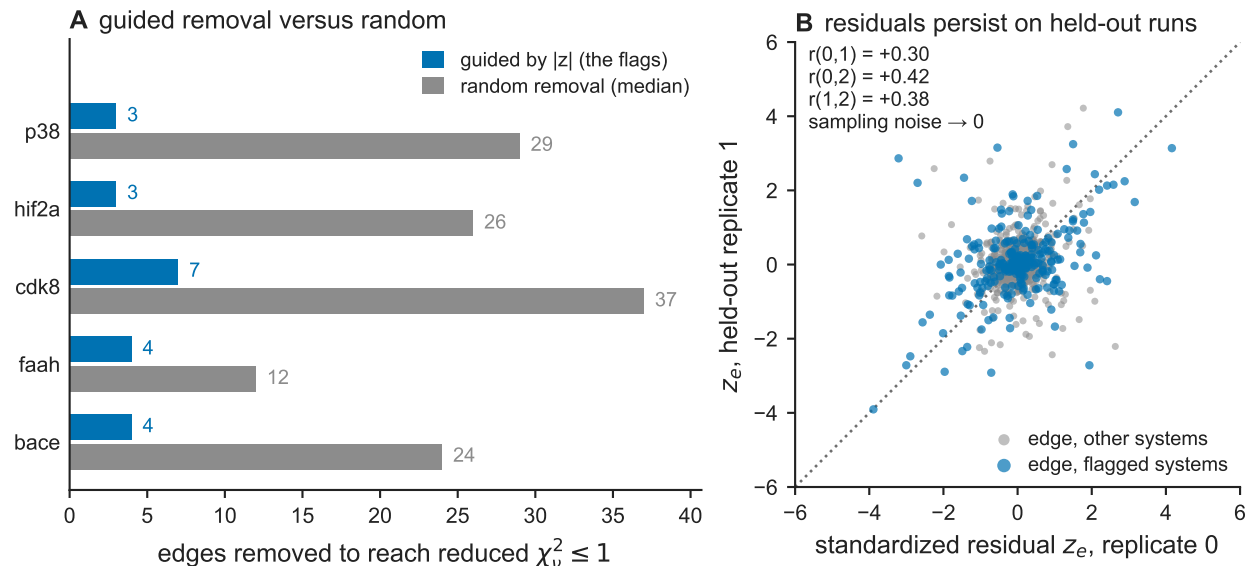


Figure 5: **Edge-removal repair test and cross-replicate residual reproduction.** (A) Number of edges removed, in descending order of $|z|$, before reduced $\chi^2 \leq 1$ is restored: guided removal of the flagged high- $|z|$ edges versus random removal, per system. (B) Standardized residual z_e from a replicate-0 network fit versus z_e from a replicate-1 fit, for edges in flagged and in non-flagged systems, with the $y = x$ line and the Pearson correlation between each pair of replicates.

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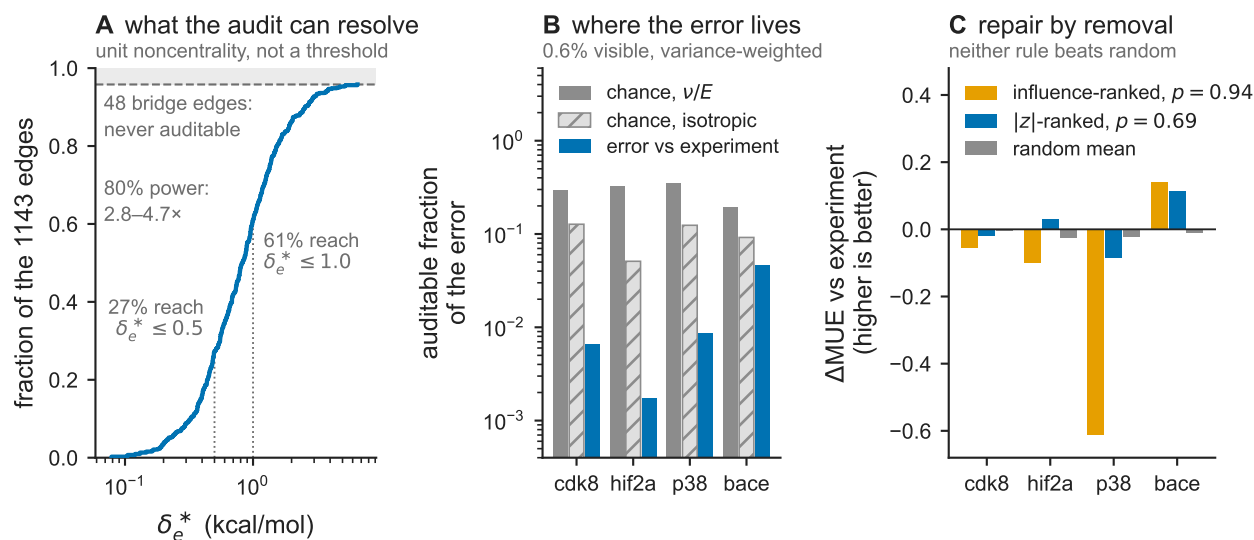


Figure 6: **Auditable capacity and the location of the error.** (A) Cumulative fraction of the 1143 benchmark edges against δ_e^* , the shift at unit noncentrality, with the bridge edges marked as the ceiling they impose. (B) Fraction of a system’s error against experiment lying in the auditable subspace, against the fraction an unaligned error field would show under two conventions: whitened-isotropic (dof/E) and isotropic in kcal/mol, for the four systems selected by the test. (C) Change in mean unsigned error against experiment after removing the same number of edges chosen by influence, by standardized residual, and at random.

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Supporting Information

Reproducibility. The analysis code released with this work regenerates every figure in this article, each from a single command, and it emits the per-system tables reported here from the same run that draws the corresponding figure, so the two cannot drift apart. This covers the quality-control results (Figures 4 and 5 of the main text and Figures S8, S9 and S10 here), the observability map of Figure S5, the fixed-cutoff comparison of Figure S7, the σ -profile sweeps of Figures S11 and S12, and the autocorrelation panel of Figure S1. The constants appearing in the theorems and the invariants of the cycle-closure test are additionally asserted as unit tests that run in the same environment. Every input is public: the alchemtest BACE1 and benzene sets, the OpenFF protein–ligand benchmark, ChEMBL [39], and the OpenFE IndustryBenchmarks2024 replicate set [4]. Recalibration baselines use temperature scaling [14] and isotonic regression [38].

The differentiable backward pass. The BAR layer’s backward pass costs one division, as quoted in the main text. No experiment in this article trains through the layer, so the result establishes a capability of the layer rather than underwriting a measurement.

Theorem S1 ($O(1)$ differentiable backward). *Let $\widehat{\Delta F}(\theta)$ be defined by $S(\widehat{\Delta F}, \theta) = 0$ with non-degenerate curvature $I \neq 0$ (positive overlap). The implicit function theorem gives $d\widehat{\Delta F}/d\theta = I^{-1}\partial_\theta S$. For window means, $d\widehat{\Delta F}/d\mu^f = I^f/I$, $d\widehat{\Delta F}/d\mu^r = -I^r/I$, with $|I^f/I| + |I^r/I| = 1$. The backward pass is one division by the curvature; the root-find is not unrolled.*

The score S , the curvature I and the unified coordinates are those of Section 3 of the main text. The released layer’s gradients are checked against finite differences to machine precision in the unit tests.

BACE1 per-window overlap values (Figure 2B). Figure 2B reports that naive $1/I$ overstates the bootstrap se by 2–12 $\times$ on the 19 real BACE1 adjacent- λ BAR windows (mean 6.07, [2.78, 12.01]), the worst overstatement occurring at high overlap, consistent with the controlled panel (invariant #1, main text). These 19 rows are the windows of one λ schedule, taken from the complex leg of the alchemtest AMBER `load_bace_example` relative-binding calculation across its decharge, recharge and van der Waals legs; each is a two-state BAR estimate between neighbouring λ values, so the set tests the closed-form variance against a decorrelated bootstrap on real molecular-dynamics works, and it does not test 19 separate binding free energies or reach beyond the high-overlap band this schedule occupies. Table S1 lists the (λ_i, λ_j) window labels, per-window overlap $4\langle p(1-p) \rangle$, and both ratios for all 19, emitted verbatim, λ labels included, by the released analysis code (seed 20260629; both sandwich inputs and the bootstrap truth subsampled to the statistical inefficiency g before B is formed, Section 3; numerically identical to the binding rows of the corresponding panel, committed with the analysis code), so the λ pairing is re-derivable from the released code. Every window sits at overlap ≥ 0.84 (mean 0.93); the worst naive/bootstrap ratio (12.01 $\times$) occurs at the single highest-overlap window ($4\langle p(1-p) \rangle = 0.992$), and the best (least overstated, 2.78 $\times$) occurs at the lowest-overlap window in the set (0.846), the same high-overlap-is-worst regime as the controlled sweep of Figure 2A rather than a property of the particular windows sampled. The trend across the full 19-window set is not confined to the extremes: the Spearman rank correlation between overlap and the naive/bootstrap ratio is $\rho = 0.84$ ($n = 19$, computed from Table S1), so the relationship holds within the narrow [0.846, 0.992] overlap band, not just at its endpoints.

Autocorrelation robustness: raw n vs $n_{\text{eff}} = n/g$ (Figure S1). The sandwich $B = n_f \text{Var}_f[p] + n_r \text{Var}_r[p]$ (Theorem 1) assumes independent samples, whereas real molecular-dynamics works are autocorrelated. On the 19 real BACE1 adjacent- λ BAR windows of Figure 2B (no new molecular dynamics), the sandwich se is recomputed at the raw sample count and again after decorrelating each leg to its statistical inefficiency g [32] (geometric-mean $\bar{g} = 1.48$, range [1.15, 2.16]; median count $n=500 \rightarrow 349$). Against the correlation-aware truth (the decorrelated bootstrap standard deviation, the same reference as Figure 2B), the n_{eff} sandwich is calibrated (reported/truth 1.00, [0.93, 1.07], reproducing Figure 2B), while the raw- n sandwich is overconfident by 0.83 $\times$ ([0.67, 0.97]), matching the $1/\sqrt{\bar{g}} = 0.82$ autocorrelation prediction. The sandwich also tracks its own matched (raw) bootstrap (1.00, [0.92, 1.05]): the closed form is internally consistent, and the only quantity that must be right is the effective sample count, which the paper decorrelates to g throughout (Section 3, paper text). The calibration verdict is therefore robust to autocorrelation once n_{eff} is used, as it is in every real-data panel.

Table S1: **Per-window overlap on the 19 real BACE1 adjacent- λ BAR windows.** λ_i, λ_j are the BAR window endpoints; $4\langle p(1-p) \rangle$ is the normalized BAR overlap; sandwich/boot and naive/boot are the reported-se/bootstrap-se ratios for the sandwich B/I^2 and the naive $1/I$, respectively, against a decorrelated-bootstrap reference standard error. Rows are sorted by overlap, ascending; repeated (λ_i, λ_j) pairs are distinct windows from BACE1’s decharge, recharge, and van der Waals alchemical legs, which share a common λ schedule.

λ_i	λ_j	$4\langle p(1-p) \rangle$	sandwich/boot	naive/boot
0.437	0.563	0.846	0.992	2.780
0.316	0.437	0.858	1.001	3.020
0.206	0.316	0.876	0.967	4.377
0.750	1.000	0.884	0.956	3.315
0.563	0.684	0.909	1.041	3.451
0.500	0.750	0.917	0.971	3.618
0.684	0.794	0.921	0.997	3.581
0.250	0.500	0.927	1.017	3.860
0.000	0.250	0.928	0.967	3.678
0.500	0.750	0.943	1.018	9.823
0.794	0.885	0.943	0.981	4.165
0.000	0.250	0.948	0.986	9.360
0.115	0.206	0.949	1.005	6.414
0.250	0.500	0.967	1.004	9.116
0.750	1.000	0.968	1.052	9.901
0.885	0.952	0.969	0.995	5.670
0.952	1.000	0.983	1.064	8.193
0.048	0.115	0.984	1.066	8.960
0.000	0.048	0.992	1.007	12.007

Corrected learned-variance foils and the data-efficiency floor (Figure 2A). Neither correction restores the head’s calibration at a realistic budget. A head trained on the β -weighted negative log-likelihood (β -NLL) [31] ($\beta=0.5$, the standard objective correction) does not recover calibration (se $\approx 0.11\times$, slightly worse than plain NLL’s $0.15\times$), and at this training seed a 5-member deep ensemble [23] reaches only the large-budget floor (se $\approx 0.20\times$, the same $\approx 5\times$ overconfidence a $20\times$ -larger-budget plain head reaches). The residual $\sim 5\times$ overconfidence of the plain, oracle and β -NLL heads is therefore a property of the per-edge parameterisation rather than of the label budget or of the Gaussian-NLL objective. The conditional variance $se(\text{overlap})$ is identifiable from single labels, and pooling single-label edges by overlap recovers the sandwich to $\sim 6\%$ at 4000 edges ($\sim 4\%$ at 40000) on the same labels the heads were given. This bears on the scope of the quality-control comparison. Placing a 6% miscalibration on the frozen scale grid of Figure S12 gives seven flagged systems of 48 against the published six, and a 6% over-estimate gives six; the test is unchanged. The collapse to 42 of 48 requires the $5\text{--}11\times$ under-estimate the trained per-edge heads produce, so the contrast is scoped to heads of that kind and is not evidence that a same-budget learned estimator must break the test.

Seed stability of the learned-variance foils (Figure 2A). These ratios are stable across independent training seeds for the plain and β -NLL heads: over 5 seeds the plain head spans $0.1229\text{--}0.2103\times$ (mean $0.1537\times$) and the β -NLL head $0.1024\text{--}0.1256\times$ (mean $0.1147\times$), so their overconfidence is a property of the estimator at this budget rather than of a single initialization. The 5-member ensemble is not stable: across the same 5 seeds, each with a disjoint 5-network member

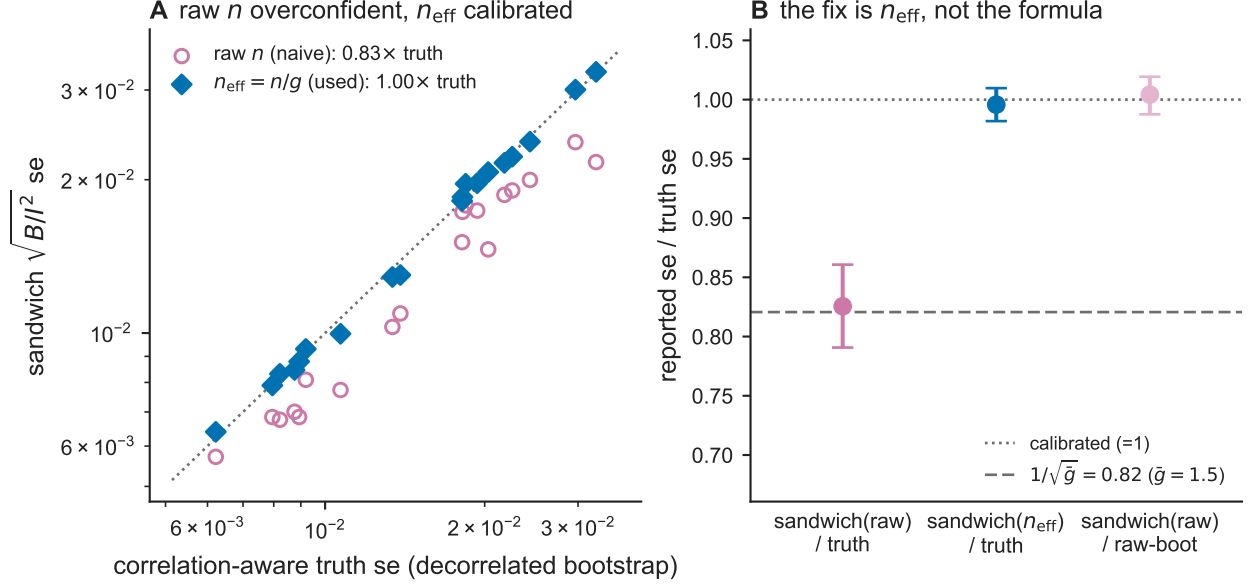


Figure S1: **Sandwich standard error at the raw sample count versus at the effective sample count $n_{\text{eff}} = n/g$, on the real BACE1 λ windows.** (A) Sandwich standard error at n_{eff} and at raw n , plotted against the correlation-aware (decorrelated bootstrap) reference standard error, with the $y=x$ line. (B) Aggregate ratio of sandwich standard error to the reference, at n_{eff} , at raw n , and of the sandwich to its own raw- n -matched bootstrap, with bootstrap confidence intervals; the $1/\sqrt{\bar{g}}$ autocorrelation prediction is marked.

block, it spans $0.1984\text{--}1.2200 \times$ (mean $0.4198 \times$, sd $0.4481 \times$), and one seed overestimates the true se by 22%, crossing perfect calibration. Unlike the plain, oracle ($0.1941\text{--}0.2571 \times$, mean $0.2101 \times$), and β -NLL heads, whose overconfidence is robust across independent retrains, the ensemble is unreliable in both directions at this budget.

Synthetic decomposition (illustration). In a controlled one-dimensional problem whose generative structure is known, the decomposed interval is up to $2.2 \times$ sharper than a split-conformal interval ($2.0 \times$ sharper than conformalized quantile regression) at equal coverage, with better conditional calibration. The problem illustrates the mechanism; the evidence on real data is Figure S14.

Edge weighting (Figure S2). The acquisition policy outperforms random selection by a factor of 2.2. Changing the edge weighting from naive to sandwich changes calls-to-top- k by a factor of 0.98 [0.93, 1.03] (24-seed bootstrap confidence interval), which is the Gauss–Markov ceiling on the choice of weights.

Gauge-aware acquisition (Figure S3). The Fisher–resistance Laplacian (Theorem 2) makes the knowledge gradient exactly zero on gauge (all-ones) contrasts. A controlled simulation gives $\text{KG} = 10^{-21}$ on the all-ones gauge contrast, an exact zero, and the identifiability statement behind it transfers structurally to multi-target networks. A gauge-unaware acquisition spends budget resolving a decision-irrelevant cross-cluster offset, reaching the within-target top- k at regret 0.27 where the gauge-aware one reaches 0.01.

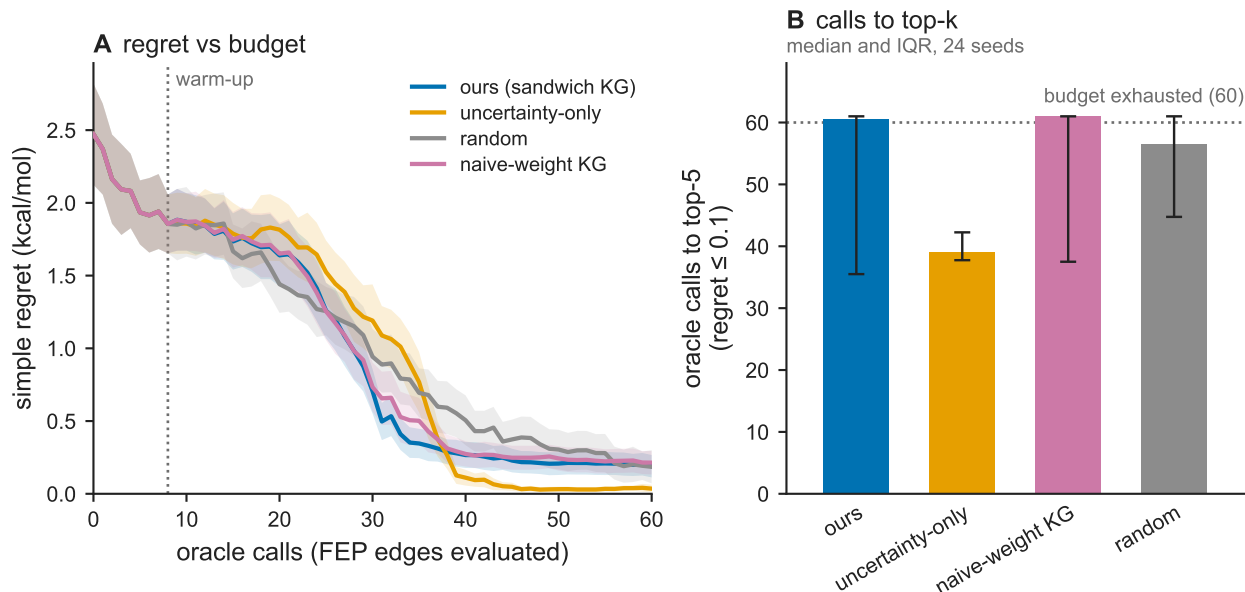


Figure S2: **Simulated active-learning acquisition policies.** (A) Simple regret versus oracle calls (simulated FEP edges evaluated), for a sandwich-weighted knowledge-gradient policy, a naive-weighted knowledge-gradient policy, an uncertainty-only policy, and random selection, averaged over 24 seeds with 95% confidence bands. (B) Median oracle calls to reach the top- k candidates (regret at or below tolerance), with interquartile-range error bars, for the same four policies.

Structural context of the flagged systems (Figure S4). Two of the systems the cycle-closure test flags (Figure 4) carry an independently-documented hard structural feature, read here directly from public co-crystal structures. In BACE1 (PDB 4DJW) the catalytic aspartic-acid dyad hydrogen-bonds the inhibitor (Asp93 and Asp289, both 2.7 Å); the protonation state of an aspartic-protease dyad is a recognized source of free-energy error. In BRD4 BD1 (PDB 3MXF, in complex with JQ1 [10]) the acetyl-lysine pocket packs 16 crystallographic waters within 4.5 Å of the ligand (8 within 3.5 Å) around the recognition asparagine Asn140 (3.1 Å); this conserved water network is the known BRD4 free-energy challenge [1]. The other flags carry analogous a-priori difficulties: *faah* (covalent serine hydrolase, large acyl channel), *cdk8* and *p38* (large scaffold and R-group changes), and *hif2a* (a buried polar cavity). The context is retrospective and structural rather than a per-edge causal claim: cycle closure is edge-level and blind to node-consistent bias (main text). It establishes only that the flagged set coincides with the features a domain expert would identify a priori. In Figure S4 the colours annotate structural objects rather than methods.

Per-edge observability map and predictive falsifier (Figure S5). The estimation-detection conservation law (Theorem 3) certifies $\sum_e h_e = \text{dof}$ to machine precision (maximum deviation 1.78×10^{-14}) across the 48 cyclic systems and identifies 48 structurally un-auditable bridge edges ($h_e = 0$, one per system on average); a pre-registered test of whether the curl-leverage also predicts the location of reproducible systematic error, run on 934 pooled edges over 45 systems against the rule confirmed if and only if $\rho > 0$ and $p < 0.05$ (10^4 within-system block permutations), is negative (studentized Spearman $\rho = -0.073$, $p = 0.9571$), so only the structural certificate is claimed.

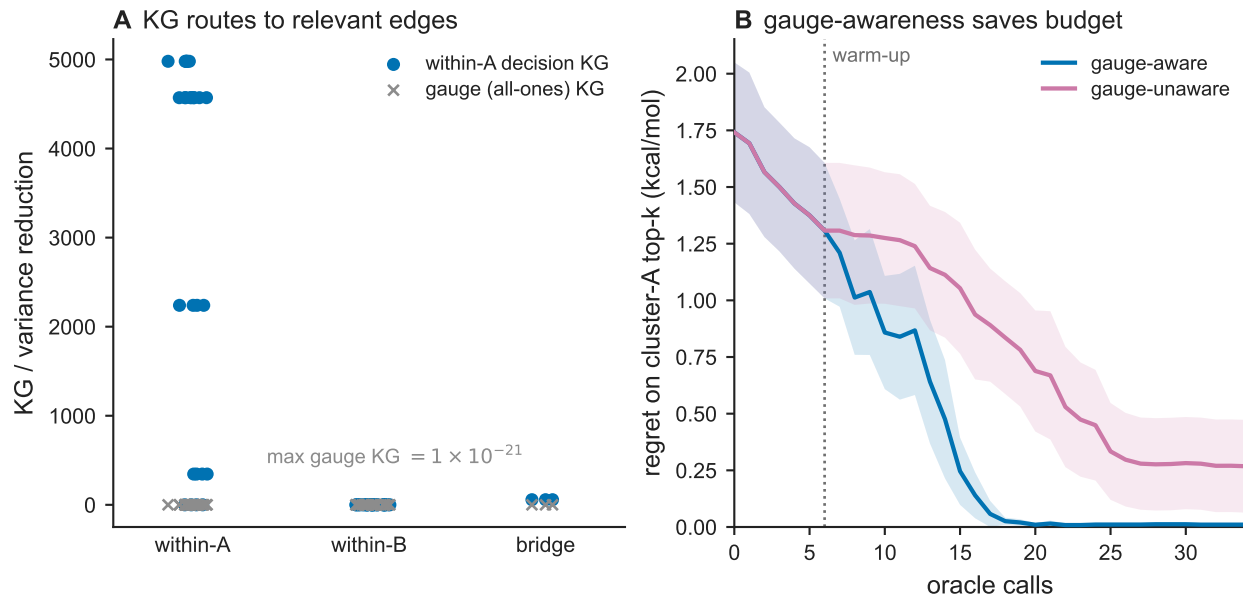


Figure S3: **Knowledge-gradient score on gauge and decision-relevant contrasts (simulation)**. (A) Knowledge gradient for the within-cluster decision contrast and for the gauge (all-ones) contrast, by edge type (within-cluster-A, within-cluster-B, bridge). (B) Regret on the cluster-A top- k candidates versus oracle calls, for a gauge-aware and a gauge-unaware acquisition policy, averaged over seeds with 95% confidence bands.

Per-system auditability (Table S3). Theorem 4 makes a network’s auditable fraction a property of its topology and its reported precisions alone, so it can be tabulated for every benchmark system before any question of what that system’s error turns out to be. Table S3 gives, per system, the ligand and edge counts, the number of independent cycles ν , the auditable fraction ν/E , the bridge count, and the median detectable shift over that system’s auditable edges. The auditable fraction is exactly $1 - (N - c)/E$, for N ligands, E edges and c connected components, and so is small wherever a network is sparse: it runs from 0.12 to 0.46 with a median of 0.31, and no benchmark system exposes more than half of its own error space to the test. The eight flagged systems span essentially that whole range, from 0.12 to 0.46, rather than concentrating at the auditable end: a system is flagged for the error it carries, not for how much of that error is in view. **brd4**, flagged on two of the three replicates, sits at the bottom of the table with a single independent cycle: its reduced χ^2 is a one-degree-of-freedom statistic, which is the quantitative form of the caution Section 4.2 places on it. Rows are ordered by auditable fraction, and the flag marks are recomputed from the three replicates by the same code that draws Figure 6, not transcribed from the main text.

What was pre-registered. Several arguments in this article turn on what was frozen before the computation rather than chosen after it. The design file governing the decomposition increment was committed to the article’s repository before the commit that computes any of the quantities it governs, and it is released with the analysis code; the Data and Software Availability statement says where. It fixes three items and no others: H1, the visible fraction of the measured error against experiment with its chance level and its success criterion; H2, the influence-ranked repair race; and H3, the auditability map, which it labels descriptive and gives neither threshold nor verdict. Both success criteria are conjunctive, so a single conjunct firing is not a success. H1 is labelled a

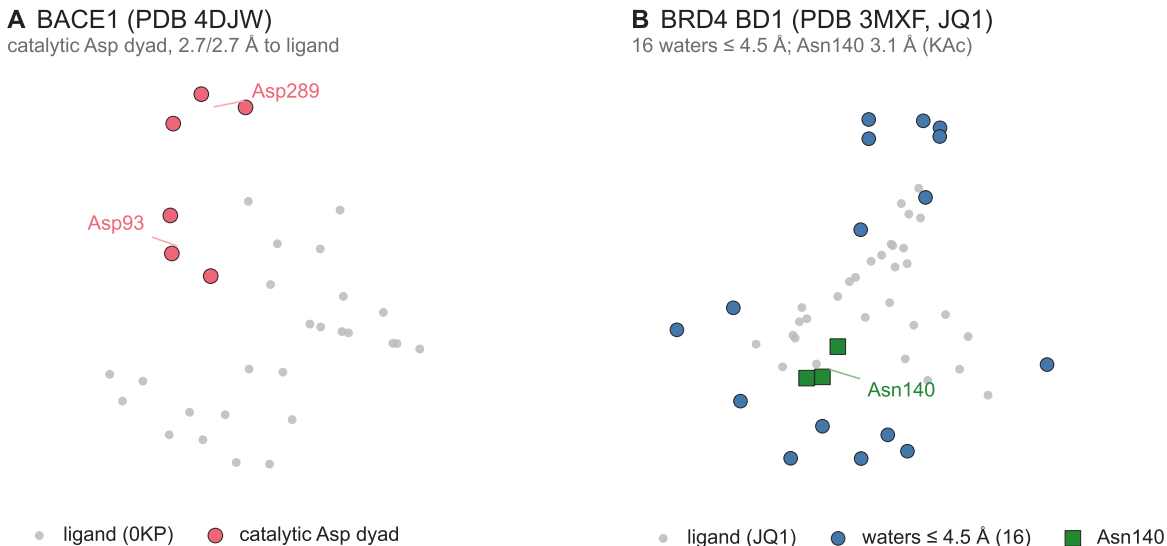


Figure S4: **Structural context of two flagged systems, from public co-crystal structures.** Heavy-atom coordinates. (A) BACE1 (PDB 4DJW): the catalytic aspartic-acid dyad (Asp93/Asp289, red circles) and the bound inhibitor (grey). (B) BRD4 BD1 (PDB 3MXF, in complex with JQ1): crystallographic waters within 4.5 Å of the ligand (blue circles), the recognition residue Asn140 (green squares), and the ligand (grey).

quantification backed by an identity rather than a risky test, and says why: experimental affinities are per-ligand, so $\Pi\epsilon$ reduces to the closure residual and the visible fraction is an identity, with only the comparison against the chance level at risk. The confound the design cannot resolve is declared in it: experimental measurement error and the stereo-blind affinity join are both per-ligand and therefore also gradient fields, so the design cannot separate per-ligand force-field bias from per-ligand experimental or annotation error, and the conclusion it licenses is the first of those and not the second. H2 is declared in the design file to be a second look at data that had already produced one null, and its result carries that status.

The other pre-specified constants this article invokes are recorded in the released analysis code and in the pre-registration for the experiment that fixed them, not in that file: the $|z|$ -rule conjunctive criterion, the curl-leverage falsifier rule, the fixed-cutoff head-to-head criterion, the 1.0 kcal/mol cutoff itself, the swept scale grid with its two frozen readouts, the per-edge granularity of the self-calibrated null, and the close-the-loop criterion. Three of them live only in code, and the paragraph below audits those three against the descriptions printed here.

Audit of the frozen criteria that live only in code. Three criteria are recorded in the analysis code and not in the design file, and each is audited here against the description printed in this article. The curl-leverage falsifier is confirmed if and only if $\rho > 0$ with a within-system block-permutation $p < 0.05$ over 10^4 permutations; the code implements exactly that and the supplement prints exactly that. The fixed-cutoff head-to-head requires the calibrated null’s detection area to exceed both baselines, implemented as a paired bootstrap of $AUC(A) - \max(AUC(B), AUC(C))$ with a win declared only when the point difference is positive and the lower confidence bound excludes zero; the printed description matches, and the tie follows from the +0.140 [−0.008, 0.316]

against the stronger baseline. The $|z|$ -rule conjunctive criterion matches the code as printed here, requiring Stouffer $p < 0.05$ and a guided ΔMUE above the 95th percentile of its own random null in a majority of grounded systems, and that is the upper tail because ΔMUE is defined as $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, so an improvement is positive. Its frozen file names the lower tail, "below the 5th percentile", which is the same event stated in the opposite sign convention; the code’s variable name follows the file rather than the statistic. No result depends on the discrepancy. The frozen file is unaltered, and the description printed in this article is the one the code implements.

The visible fraction is metric-dependent (Section 4.1). The auditable share of the error against experiment is a ratio of squared norms, so it depends on the metric those norms are taken in, and the main text quotes it in the variance-weighted metric the closure statistic itself uses. Three readings of the same four sub-networks are collected in Table S2, each with its own chance level and with the range the fraction takes when one edge is deleted at a time. The whitened reading takes both quantities in the variance-weighted metric the closure statistic lives in, against dof/E . The isotropic reading keeps that fraction and measures it against $\text{tr}(\Pi W)/\text{tr}(W)$, the chance level an error isotropic in kcal/mol would produce under the same projector; it gives the weakest margins, and it is the reading the abstract and the main text quote. The unweighted reading recomputes the projector itself in the unweighted metric, $\Pi = I - A(A^\top A)^+ A^\top$ on the raw incidence matrix A , so neither the fraction nor its chance level carries any variance weighting; there the pooled fraction rises from 0.6% to 5.45%. The worst single cell in the whole table is **bace** under the isotropic reading with one edge deleted, at a factor of 1.36 below its chance level. Both the fraction and the chance level are recomputed on every deletion, since deleting an edge moves the projector as well; **make_figHodge** produces the table. **cdk8**’s flag is a cross-release join rather than a statement about either release’s simulations (Section 4), so the pooled figure is given both ways: 0.0060 whitened and 0.0545 unweighted over the four systems, 0.0060 and 0.0610 with **cdk8** dropped, so the whitened headline does not depend on it. Recomputing the ratio on medians rather than sums, which removes any dependence on a few large residuals, gives 0.0034, 0.0068, 0.0052 and 0.0015, the same order throughout; and **bace**, the one grounded system whose ligand set contains no enantiomer pair collapsed by the stereo-blind affinity join, shows the effect as plainly as the three that do. Every reading puts the auditable share below its own chance level; what changes is by how much, and the conclusion the main text draws is the weakest of them.

The label source can also be changed. The measurement uses a stereo-blind, target-wide ChEMBL search under the coverage rule, and the benchmark of Section 5 carries curated per-edge experimental $\Delta\Delta G$ for three of the four systems (**bace** has none). Matching those to the network by canonical SMILES recovers 57%, 98% and 91% of the curated edges for **cdk8**, **hif2a** and **p38**. The matched sub-networks are much smaller than the grounded ones and **p38**’s retains no cycle, so the check is weak, but where it can be run it agrees: the visible fraction on curated labels is 0.0007 for **cdk8** against a chance level of 0.071, and 0.0006 for **hif2a** against 0.147, in both cases at or below what the ChEMBL labels give on the same edges. Lifting the pooled fraction to its chance level by annotation noise alone would require that noise to carry almost all of the total squared standardized error, which these agreements make implausible; **bace**’s margin is the exception and is why that number does not appear in the abstract.

The system column carries the benchmark’s own name for a target’s edge network, which is not a protein: the 49 names span 41 protein labels, **bace** contributing four and **hsp90** four, so a count of flagged systems is not a count of flagged targets. Nor is a name always one released data set: eight of them combine two or more, which the paragraph below takes up. **ciordia_retro** is released under the **asap_mix** group, and the release’s own statement is the only target assignment

Table S2: **The visible fraction under three metric conventions.** For each grounded system: edges E , independent cycles ν , the auditable share of that system’s error against experiment, the chance level it is measured against, their ratio, the range the fraction takes when one edge is deleted at a time, and the worst margin over those deletions with both the fraction and the chance level recomputed on each. *Whitened* takes both in the variance-weighted metric the closure statistic lives in, against ν/E . *Isotropic* keeps that fraction and measures it against the chance level an error isotropic in kcal/mol would produce under the same projector, $\text{tr}(\Pi W)/\text{tr}(W)$. *Unweighted* recomputes the projector in the unweighted metric, so neither the fraction nor its ν/E chance level carries any variance weighting. Every cell of every convention, and every single-edge deletion, sits below its own chance level; the weakest is **bace** isotropic under deletion. Pooled over the four systems the fraction is 0.0060 whitened and 0.0545 unweighted, and 0.0060 and 0.0610 with **cdk8** dropped. Produced by `make figHodge`.

system	E	ν	metric	visible	chance	margin	leave-one-out	worst
cdk8	31	9	whitened	0.0065	0.2903	44.8×	[0.0034,0.0113]	23.60×
			isotropic, kcal/mol	0.0065	0.1273	19.6×	[0.0034,0.0113]	7.36×
			unweighted	0.0286	0.2903	10.2×	[0.0178,0.0324]	8.22×
hif2a	59	19	whitened	0.0017	0.3220	187.2×	[0.0010,0.0029]	106.60×
			isotropic, kcal/mol	0.0017	0.0510	29.6×	[0.0010,0.0029]	13.21×
			unweighted	0.0172	0.3220	18.7×	[0.0127,0.0188]	16.53×
p38	51	18	whitened	0.0086	0.3529	41.0×	[0.0033,0.0132]	25.81×
			isotropic, kcal/mol	0.0086	0.1240	14.4×	[0.0033,0.0132]	1.46×
			unweighted	0.1371	0.3529	2.6×	[0.0285,0.1499]	2.31×
bace	26	5	whitened	0.0460	0.1923	4.2×	[0.0027,0.0628]	2.55×
			isotropic, kcal/mol	0.0460	0.0919	2.0×	[0.0027,0.0628]	1.36×
			unweighted	0.0564	0.1923	3.4×	[0.0033,0.0707]	2.26×

Table S3: **What each benchmark system can be audited for.** For the 48 systems carrying at least one independent cycle: N ligands, E edges, c connected components, ν independent cycles, the auditable fraction ν/E of that system’s error space, the number of bridge edges (br), which lie in no cycle and carry no evidence at any magnitude, and the median shift $\tilde{\delta}^*$ in kcal/mol at which the closure statistic carries unit noncentrality, over that system’s auditable edges. Rows are ordered by auditable fraction ascending. A dagger marks the eight systems the Benjamini–Hochberg test flags on at least one of the three replicates. Values are replicate-0 and are produced by the same run of the released analysis code that draws Figure 6. Note that $\tilde{\delta}^*$ is not a detection threshold: reaching 80% power at $\alpha = 0.05$ needs a shift of 2.8 to 4.7 times it, and it is carried on the raw reported standard errors.

system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$	system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$
brd4 [†]	8	8	1	1	0.12	0	0.49	mc11	54	76	2	24	0.32	1	0.91
taf12	8	8	1	1	0.12	0	0.48	hif2a [†]	41	59	1	19	0.32	1	0.78
scyt_dehyd	7	7	1	1	0.14	0	0.43	shp2	26	37	1	12	0.32	2	1.14
chk1	13	15	1	3	0.20	1	1.29	syk	46	67	1	22	0.33	5	1.25
t4_lysozyme	12	14	1	3	0.21	0	0.39	bace1	3	3	1	1	0.33	0	0.64
hsp90_kung	11	13	1	3	0.23	0	0.56	dlk	5	6	1	2	0.33	0	2.09
liga	11	13	1	3	0.23	1	0.95	factor_xa	3	3	1	1	0.33	0	2.46
jnk1	24	29	2	7	0.24	3	0.93	ptp1b	26	36	2	12	0.33	5	1.39
ephx2	4	4	1	1	0.25	0	3.85	renin [†]	29	42	1	14	0.33	0	0.64
hsp90_single_ring	7	8	1	2	0.25	0	0.53	thrombin	37	54	2	19	0.35	0	0.55
hsp90_woodhead	4	4	1	1	0.25	1	0.37	keranen_p2	12	17	1	6	0.35	0	1.27
irak4_s3	4	4	1	1	0.25	1	1.60	jak2_set1	10	14	1	5	0.36	1	0.74
urokinase	4	4	1	1	0.25	0	0.41	p38 [†]	40	60	2	22	0.37	5	1.02
tnks2	27	35	1	9	0.26	0	0.47	hiv1_protease	13	19	1	7	0.37	0	0.46
faah [†]	24	31	1	8	0.26	3	0.88	cdk2	19	27	2	10	0.37	0	0.92
bace_ciordia_prospective	9	11	1	3	0.27	0	0.68	eg5	28	43	1	16	0.37	3	1.51
bace [†]	36	49	1	14	0.29	2	0.57	btb	6	8	1	3	0.38	0	1.63
bace_p3_arg368_in	21	28	1	8	0.29	8	0.71	tyk2	19	29	1	11	0.38	1	0.68
hsp90_2rings	6	7	1	2	0.29	0	0.13	itk	4	5	1	2	0.40	0	1.24
jak1	6	7	1	2	0.29	1	1.60	cmet	24	39	1	16	0.41	0	1.12
mup1	6	7	1	2	0.29	0	0.28	jak2_set2	8	12	1	5	0.42	0	1.71
hne	17	23	1	7	0.30	0	1.25	egfr	5	7	1	3	0.43	0	2.90
galectin	26	36	1	11	0.31	0	0.61	irak4_s2	5	7	1	3	0.43	0	0.98
ciordia_retro [†]	32	45	1	14	0.31	0	0.68	cdk8 [†]	35	63	1	29	0.46	3	0.98

used for it.

The component column records the number of connected pieces of a system’s edge network. Six of the 48 systems are disconnected (**cdk2**, **jnk1**, **mc11**, **p38**, **ptp1b**, **thrombin**), each into two pieces, so the components sum to 54: the benchmark’s edge lists join congeneric series that share a target without joining them to each other. Nothing in the closure statistic is affected, since dof is computed as E minus the incidence rank throughout and that rank already accounts for the extra kernel direction. Elsewhere disconnection does matter. The offset that Theorem 4(iv) says the first anchor removes is one offset per component, so a disconnected system needs one anchor in each piece before any further anchor resolves a dimension. The grounded networks of Section 4.1 inherit the same structure, **cdk8** and **bace** being disconnected once restricted to ligands carrying affinities. Their mean unsigned error against experiment is reported after a single global mean alignment, which is the conservative convention: aligning each component to its own experimental mean instead, which uses more experimental information than a prediction has access to, lowers the error by 0.067 and 0.014 kcal/mol respectively and leaves the two connected systems unchanged.

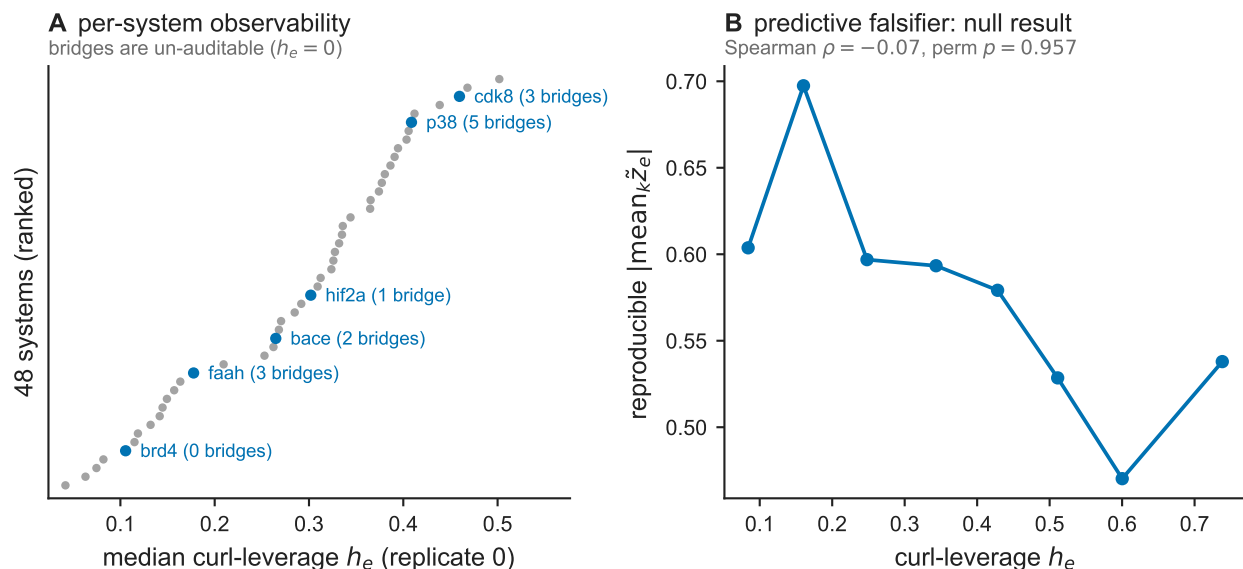


Figure S5: **Per-system curl-leverage, and curl-leverage versus reproducible residual magnitude.** (A) Median curl-leverage h_e per system, ranked, across the 48 systems with at least one independent cycle; the six cycle-closure-flagged systems are labeled with their number of bridge ($h_e = 0$) edges. (B) Curl-leverage h_e , binned into 8 quantiles, versus the mean studentized cross-replicate reproducible residual magnitude in each bin, with the Spearman correlation and permutation p -value.

Close-the-loop accuracy comparison (Figure S6). On the four flagged systems groundable against public ChEMBL affinity, guided removal of the QC-flagged high- $|z|$ edges ties a size-matched random removal on MUE-vs-experiment (combined Stouffer $p = 0.48$). Removing the highest- $|z|$ edges lowers the closure statistic (Figure 5A), but that is arithmetic and not evidence, because the statistic is the sum of exactly the per-edge quantities being removed in descending order (Section 4). The comparison adds that the same removal yields no improvement against experiment. The reach demonstrated for the quality-control test therefore stops at internal edge-level consistency and does not extend to force-field-limited absolute accuracy, the same scope boundary as the clean **eg5** system. Throughout, ΔMUE is oriented as $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, the same orientation as Figure 6C.

Fixed-cutoff head-to-head and χ^2 reconciliation (Figure S7). Two pre-registered checks were run on the same 1143 edges of the 48 cyclic systems. The first is the head-to-head: the calibrated per-edge null, scored by area under the receiver-operating-characteristic curve (AUC) against the reproducible-systematic magnitude defined in Computational details, scores 0.865 against a fixed 1.0 kcal/mol hysteresis cutoff's 0.575 (paired bootstrap difference $+0.290$ [$+0.094$, $+0.490$]) and against a fixed-se χ^2 test's 0.725 ($+0.140$ [-0.008 , $+0.316$], straddling zero). The pre-registered criterion required the calibrated null to exceed both, so the verdict is a tie. The ordering within that tie is not a discrimination gain over the fixed cutoff in standard use: the scored label standardizes each residual by the same per-edge reported variance the calibrated rule weights by, so the AUC ordering is monotone in a rule's proximity to that variance and the comparison ranks nothing. The second check tested whether correlation among residuals of edges sharing a ligand endpoint explains the apparent gap between a closure-implied se-conservatism ($1.71\times$, obtained as $1/\sqrt{0.34}$ from the median reduced χ^2) and the $1.41\times$ measured against seed and velocity repeats. It does

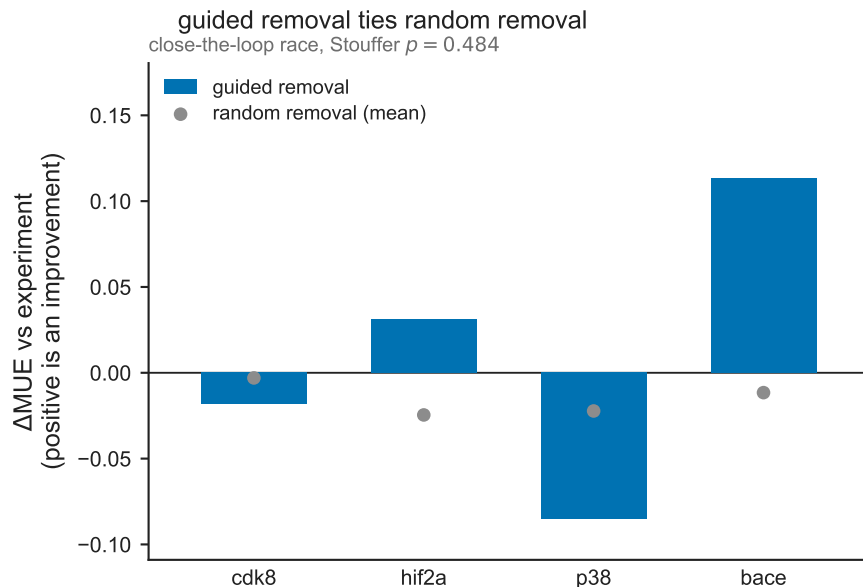


Figure S6: **Change in mean unsigned error against experimental affinity after guided versus random QC-edge removal.** Bars: change in mean unsigned error against experiment (ΔMUE) after guided removal of the QC-flagged high- $|z|$ edges, per system. Points: mean ΔMUE after a size-matched random removal, per system. Throughout, ΔMUE is $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, so a positive value is an improvement. Systems: `cdk8`, `hif2a`, `p38`, `bace`.

not: the excess correlation is -0.0079 $[-0.0314, +0.0164]$, and the Benjamini–Hochberg flag set is unchanged. The two closure-implied factors quoted in this article are the same functional on two populations and do not conflict. The pre-registered constant $1.71\times$ is $1/\sqrt{0.34}$ from the median reduced χ^2 over all 48 cyclic systems, the population of Figure 4; the $1.67\times$ annotated inside panel B is $1/\sqrt{0.361}$ over the 46 systems this second check admits, since estimating the shared-node null needs at least one disjoint edge pair per system and `bace1` and `factor_xa` are three-edge triangles that have none. Dropping those two raises the median. Neither the pre-registered constant nor the factor annotated in panel B is a conservatism estimate, for the following reason. The premise of the second check is unsound: by Theorem 3 the closure statistic identifies the curl-leverage-weighted mean of c_e^{-2} rather than a median, so $1/\sqrt{\text{median}}$ does not estimate the same functional as the replicate ratio. There is accordingly no physical gap for shared-node correlation to explain. The pre-registered constants are retained as the pre-registered record, and the verdict above is reported as run and not revised. On replicate 0 the calibrated null flags 6 of 48 systems, the fixed-se χ^2 test 11, and the fixed cutoff 26, and the three flag sets are strictly nested (calibrated $\subset$ fixed-se $\subset$ fixed cutoff), so on that replicate the calibrated null catches no system the fixed cutoff misses and instead removes 20 of its flags. The strict chain is a property of that replicate and not of the benchmark. On replicate 1 the counts are 6, 13 and 24 and two links fail: `ciordia_retro` is flagged by the calibrated null and not by the fixed-se test, and `mup1` by the fixed-se test and not by the cutoff. On replicate 2 the counts are 3, 11 and 24 and the chain holds. The calibrated set lies inside the cutoff’s on all three, which is the only containment any claim here rests on. Under the release’s own blocks rather than the by-target grouping, the same replicate-0 test over 56 testable blocks gives 6, 10 and 29, the chain nests, and the calibrated null removes 23 of the cutoff’s flags. Whether that removal is an improvement is what the head-to-head above cannot determine, for the reason given there. The 1.0 kcal/mol cutoff was pre-registered at the field-standard value before this comparison was

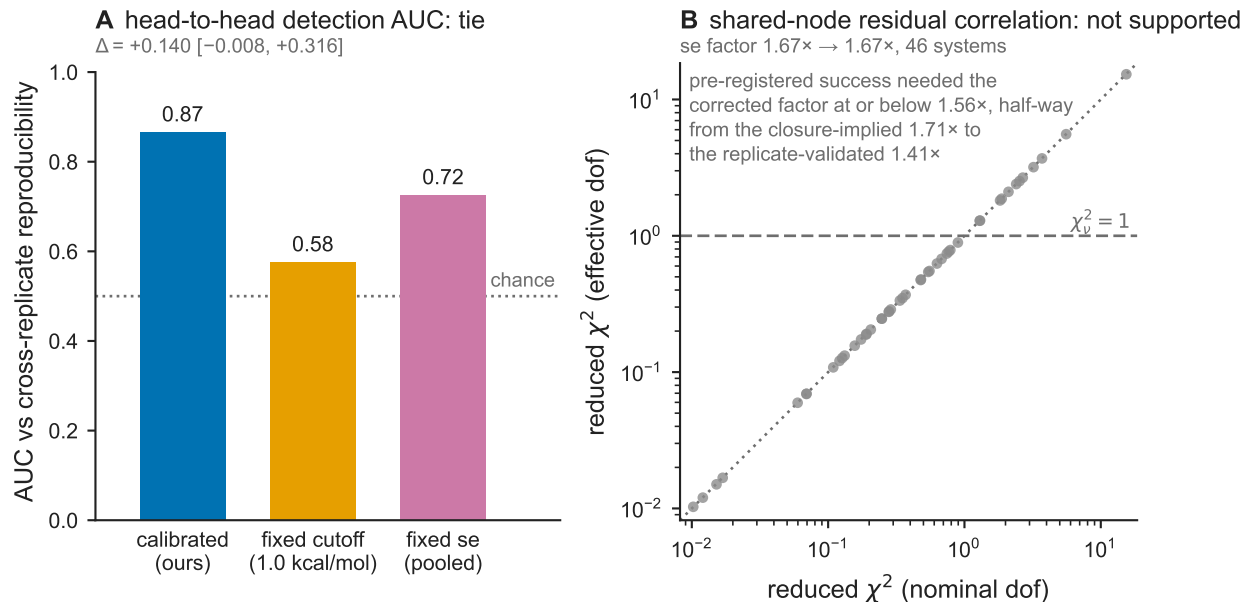


Figure S7: **Detection AUC of three flagging rules, and nominal versus correlation-corrected reduced χ^2 .** (A) Area under the ROC curve (AUC) scored against the per-system reproducible-systematic magnitude defined in Computational details, for the calibrated per-edge null, a fixed 1.0 kcal/mol hysteresis cutoff, and a fixed-standard-error χ^2 test, with the chance level (AUC = 0.5) marked. (B) Per-system reduced χ^2 at the nominal degrees of freedom versus at the shared-node-correlation-corrected effective degrees of freedom, with the $y=x$ line, the $\chi^2_\nu = 1$ reference, and the annotated closure-implied standard-error factor of $1.67\times$, the pre-registered functional evaluated over this panel’s 46 systems.

run, and no sweep over alternative thresholds was performed, so the comparison above is specific to that one fixed rule.

The flag sets under arbitrary dependence (Figure S8). The correction reported throughout is Benjamini–Hochberg, whose guarantee needs a dependence assumption between systems. Benjamini–Yekutieli [6] needs none, at the cost of running the same step-up rule at α/H_m with $H_{48} = 4.4588$, that is at 0.0112 rather than 0.05. It returns {bace, brd4, cdk8, faah, hif2a, p38} on replicate 0, identical to the published set; {bace, cdk8, renin} on replicate 2, again identical; and on replicate 1 {bace, ciordia_retro, hif2a, p38, renin}, which drops brd4 from the Benjamini–Hochberg set of six. The published set therefore does not depend on the dependence assumption, and the variation of the flag set across replicates is unchanged by removing it. `make figStab` emits the three sets.

Per-system test statistics behind the flag sets (Tables S4 and S5). The flag sets summarized in Figure S8 and in Section 4.2 are thresholded Benjamini–Hochberg adjusted p -values, so the underlying continuous quantities are reported in full rather than only their dichotomization. For each system, a generalized-least-squares cycle-closure fit under the calibrated per-edge null gives a χ^2 on ν degrees of freedom, ν being the number of independent cycles in that system’s edge network; the reduced value $\chi^2_\nu = \chi^2/\nu$ and the raw upper-tail probability follow, and those raw p -values are corrected for testing 48 systems by the Benjamini–Hochberg procedure at a false-discovery rate

Table S4: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q -values, per replicate (part 1 of 2).** Reduced χ^2 and adjusted q -value from the calibrated cycle-closure test at a false-discovery rate of 0.05, computed independently on each of the three protocol replicates, for the 48 systems with at least one independent cycle; ν is the number of independent cycles. Bold marks $q \leq 0.05$; the final column lists the replicates on which the system is flagged, and a dash where it never is. Edge and cycle counts are the replicate-0 values. Rows are ordered by replicate-0 q ascending. The remaining 24 systems continue in Table S5.

system	edges	ν	replicate 0		replicate 1		replicate 2		flagged
			χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	
bace	49	14	5.57	3.2×10^{-9}	7.33	7.0×10^{-14}	7.10	2.9×10^{-13}	0, 1, 2
cdk8	63	29	2.68	6.1×10^{-5}	1.45	2.8×10^{-1}	2.62	1.1×10^{-4}	0, 2
brd4	8	1	15.40	1.4×10^{-3}	8.05	3.6×10^{-2}	4.94	1.7×10^{-1}	0, 1
faah	31	8	3.71	2.0×10^{-3}	0.45	1.0×10^0	1.47	7.1×10^{-1}	0
hif2a	59	19	2.53	2.0×10^{-3}	2.96	2.4×10^{-4}	2.05	5.4×10^{-2}	0, 1
p38	60	22	2.40	2.0×10^{-3}	2.48	1.7×10^{-3}	1.81	1.1×10^{-1}	0, 1
mc11	76	24	1.82	5.5×10^{-2}	1.79	6.8×10^{-2}	1.04	1.0×10^0	–
bace_p3_arg368_in	28	8	1.88	3.5×10^{-1}	1.88	2.8×10^{-1}	2.32	1.4×10^{-1}	–
taf12	8	1	3.21	3.9×10^{-1}	0.15	1.0×10^0	0.33	1.0×10^0	–
liga	13	3	2.10	4.7×10^{-1}	0.76	1.0×10^0	1.80	6.9×10^{-1}	–
thrombin	54	19	1.30	7.4×10^{-1}	0.86	1.0×10^0	0.61	1.0×10^0	–
renin	42	14	1.29	8.1×10^{-1}	2.81	3.1×10^{-3}	3.32	3.8×10^{-4}	1, 2
ciordia_retro	45	14	0.89	1.0×10^0	3.86	3.1×10^{-5}	0.40	1.0×10^0	1
tyk2	29	11	0.79	1.0×10^0	1.06	1.0×10^0	0.78	1.0×10^0	–
ephx2	4	1	0.77	1.0×10^0	1.67	7.7×10^{-1}	1.33	9.3×10^{-1}	–
shp2	37	12	0.74	1.0×10^0	1.21	8.7×10^{-1}	0.64	1.0×10^0	–
itk	5	2	0.68	1.0×10^0	0.08	1.0×10^0	0.10	1.0×10^0	–
tnks2	35	9	0.63	1.0×10^0	0.43	1.0×10^0	2.07	1.7×10^{-1}	–
mup1	7	2	0.55	1.0×10^0	3.05	2.8×10^{-1}	0.29	1.0×10^0	–
scyt_dehyd	7	1	0.54	1.0×10^0	1.34	8.5×10^{-1}	1.23	9.3×10^{-1}	–
cdk2	27	10	0.48	1.0×10^0	1.33	7.7×10^{-1}	1.22	9.3×10^{-1}	–
eg5	43	16	0.48	1.0×10^0	0.40	1.0×10^0	0.30	1.0×10^0	–
galectin	36	11	0.37	1.0×10^0	0.22	1.0×10^0	0.88	1.0×10^0	–
jak2_set1	14	5	0.35	1.0×10^0	0.83	1.0×10^0	0.15	1.0×10^0	–

of 0.05, giving the adjusted q -values tabulated below. The $q \leq 0.05$ rule reproduces the released implementation exactly, and that correspondence is asserted on every set the analysis emits. Each of the benchmark’s three protocol replicates is analysed independently: the fit is rerun from scratch on that replicate’s edges and the correction is applied within that replicate alone, so the three q columns are not a joint error rate across replicates. All values in both parts of the table are produced by the same run of the released analysis code that draws Figure S8.

Out-of-sample behaviour of the calibrated null (Figure S9). Each candidate held-out statistic is calibrated against a no-error null before use, and the realized level of every candidate is reported. Two of the candidates fire far above nominal and are not used; the remaining two hold their nominal level and are the ones carried into the selection comparisons.

Two-sided self-calibration of the per-edge error bar (Figure S10). Seven arms are compared: the nominal test, which uses the reported standard errors unchanged; the two-sided correction at the pre-specified per-edge granularity, which corrects each edge by its own ratio; a coarser variant correcting each system by its aggregate ratio; a variant clipping the ratio to $[1/3, 3]$; a variant divid-

Table S5: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q -values, per replicate (part 2 of 2).** Table S4 continued: the remaining 24 of the 48 systems, in the same order and with the same conventions. No system in this part is flagged on any replicate.

system	edges	ν	replicate 0		replicate 1		replicate 2		flagged
			χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	
hsp90_2rings	7	2	0.34	1.0×10^0	0.29	1.0×10^0	0.12	1.0×10^0	–
factor_xa	3	1	0.31	1.0×10^0	0.01	1.0×10^0	0.02	1.0×10^0	–
keranen_p2	17	6	0.29	1.0×10^0	0.19	1.0×10^0	0.04	1.0×10^0	–
chk1	15	3	0.28	1.0×10^0	0.27	1.0×10^0	0.69	1.0×10^0	–
hiv1_protease	19	7	0.28	1.0×10^0	0.43	1.0×10^0	0.94	1.0×10^0	–
irak4_s2	7	3	0.25	1.0×10^0	0.15	1.0×10^0	0.04	1.0×10^0	–
t4_lysozyme	14	3	0.25	1.0×10^0	0.57	1.0×10^0	2.14	5.0×10^{-1}	–
ptp1b	36	12	0.21	1.0×10^0	0.20	1.0×10^0	0.58	1.0×10^0	–
btk	8	3	0.19	1.0×10^0	2.15	4.0×10^{-1}	1.19	9.3×10^{-1}	–
hsp90_kung	13	3	0.19	1.0×10^0	0.18	1.0×10^0	0.83	1.0×10^0	–
cmet	39	16	0.17	1.0×10^0	0.36	1.0×10^0	0.42	1.0×10^0	–
hsp90_single_ring	8	2	0.16	1.0×10^0	0.02	1.0×10^0	0.24	1.0×10^0	–
hne	23	7	0.13	1.0×10^0	0.28	1.0×10^0	0.07	1.0×10^0	–
urokinase	4	1	0.13	1.0×10^0	0.01	1.0×10^0	0.25	1.0×10^0	–
jnk1	29	7	0.12	1.0×10^0	0.12	1.0×10^0	1.19	9.3×10^{-1}	–
syk	67	22	0.11	1.0×10^0	0.20	1.0×10^0	0.25	1.0×10^0	–
egfr	7	3	0.07	1.0×10^0	0.01	1.0×10^0	0.02	1.0×10^0	–
irak4_s3	4	1	0.07	1.0×10^0	0.07	1.0×10^0	0.05	1.0×10^0	–
jak2_set2	12	5	0.06	1.0×10^0	1.02	1.0×10^0	0.49	1.0×10^0	–
bace1	3	1	0.05	1.0×10^0	1.09	8.9×10^{-1}	0.43	1.0×10^0	–
dlk	6	2	0.02	1.0×10^0	0.06	1.0×10^0	0.05	1.0×10^0	–
jak1	7	2	0.02	1.0×10^0	0.32	1.0×10^0	0.10	1.0×10^0	–
bace_ciordia_prospective	11	3	0.01	1.0×10^0	0.07	1.0×10^0	0.04	1.0×10^0	–
hsp90_woodhead	4	1	0.01	1.0×10^0	0.07	1.0×10^0	0.15	1.0×10^0	–

ing the replicate standard deviation by the constant $c_4(3) = 0.886$, the expectation of the sample standard deviation at three draws, which widens the bars; a variant measuring the ratio on the two replicates the test does not read; and, at the same per-edge granularity, the one-sided correction that inflates only where the measurement says the bar is too tight, which is the direction of Table S7 applied per edge rather than per system. The constant $c_4(3)$ is unrelated to the per-edge calibration ratio c_e . Table S6 tabulates the four granularity-by-direction combinations.

Simulation protocols. The two protocols the main text names are specified here.

The pure-null simulation behind the realized levels. Every realized level quoted for a self-calibrated arm, in Section 4, in Section 7 and in the last column of Supplementary Table S6, is measured on a synthetic benchmark that carries no systematic error, so that every flag it produces is a false positive by construction. The topologies are the real ones: the same 48 systems the closure test admits, each with its own ligand-pair edge list and its own admission by the same rule (≥ 3 edges, ≥ 1 independent cycle), which depends on topology alone. The variances are the real ones: each edge keeps the per-edge reported standard error it carries, and that value is then declared to be the edge’s true sampling standard deviation, which makes the reported bars perfectly calibrated by construction. The draw law is the null: every node potential is exact, so each edge’s true $\Delta\Delta G$ is exactly the difference of its endpoints’ potentials, and an observation is that value plus an independent Gaussian of the edge’s own standard deviation. There is no systematic component anywhere and no correlation between edges. The replicate structure mirrors the benchmark’s: three independent draws per edge, of which the first plays the role of replicate 0 and is the one the closure statistic is computed on, while all three are read by the arm’s own calibration formula exactly as that arm reads the real replicates (the held-out arm reads only the second and third). Each arm’s correction, tail probability and Benjamini–Hochberg step-up at $\alpha = 0.05$ are then the identical code paths used on the real data. The simulation is run for 400 draws, so a realized probability is measured to a Monte-Carlo standard error of at most $\sqrt{0.25/400} = 0.025$, and the generator is seeded once for the whole sweep, which makes the reported levels reproducible.

The reproducible-systematic magnitude used as ground truth. Detection comparisons need a label that does not come from the detector being scored, and the only one these data supply is reproduction across the three protocol repeats. It is defined as follows. Each replicate’s network is fitted separately by the same generalized least squares, giving for replicate k and edge e a standardized residual $z_e^{(k)} = r_e^{(k)} / \sqrt{V_e^{\text{rep}}}$. The three signed residuals of an edge are averaged, $\bar{z}_e = \frac{1}{3} \sum_k z_e^{(k)}$, so that a residual reproducing in sign and magnitude survives the average while sampling noise cancels toward zero, and the edge’s score is $|\bar{z}_e|$. A system’s score is the median of $|\bar{z}_e|$ over its edges. This is the quantity the detectors of the head-to-head comparison are ranked against, by Mann–Whitney AUC of the binary flag against the per-system score. It uses only replicate residuals and no detector output, but it is not independent of the rules being scored, and the dependence is the reason no ordering is read out of that comparison (Section 4): $z_e^{(k)}$ divides by V_e^{rep} , the same per-edge reported variance the calibrated rule uses as its weight, so the label rewards a rule for standing close to that variance. “Edges whose sign and magnitude reproduce across independent replicates”, wherever that phrase appears above, means edges with a large $|\bar{z}_e|$ in this sense.

Model architectures. The two learned models the main text compares against or uses as an amortized predictor are specified here in full.

Amortized predictor. The predictor used in the acquisition, stopping and reward experiments, referred to elsewhere in this article as the trunk, is small, and it is not an equivariant network. A ligand is featurized as a 1024-bit ECFP4 Morgan fingerprint (radius 2, chirality flag off) concatenated with six achiral RDKit descriptors: molecular weight, Crippen log P , topological polar surface area, and counts of hydrogen-bond donors, hydrogen-bond acceptors and rotatable bonds. An edge is represented by the difference of its two endpoint feature vectors. A single parity-odd coordinate is appended, the difference of a summed signed volume over the assigned stereocentres of an ETKDv3 embedding, giving 1031 features per edge; every other coordinate is invariant under reflection, so a readout without it would be constant on an enantiomer pair. The regressor is a bootstrap deep ensemble of eight multilayer perceptrons, each with two hidden layers of 64 units and SiLU activations and a scalar output, trained for 300 epochs on a squared-error objective with Adam at learning rate 5×10^{-3} and weight decay 10^{-4} , on inputs standardized by the training mean and standard deviation. Each member is seeded independently and fitted to its own bootstrap re-sample; the epistemic term is the spread across members, and the total σ combines it in quadrature with the aleatoric floor before a conformal rescaling.

Learned-variance foil. The learned head the calibrated bar is compared against is a separate and much smaller model from the amortized predictor above, and it is trained on the controlled panel rather than on the benchmark. Each training example is one simulated edge, from which six features are formed: the means and standard deviations of the forward and reverse work samples, the absolute difference of the two means, and the normalized overlap scalar. Including the overlap scalar gives the head the same overlap information the closed form uses, so the comparison is fair. The network is a multilayer perceptron with two hidden layers of 32 units and SiLU activations and two outputs, a mean and a log-variance, trained for 1500 steps of Adam at learning rate 5×10^{-3} on a Gaussian negative log-likelihood, with inputs standardized by the training mean and standard deviation. The realistic-budget arm trains on 200 edges of 20 work samples each; the large-budget arm raises that to 4000 edges, twenty times larger, and the corrected arms replace the objective by the β -NLL of Seitzer et al. [31] at $\beta = 0.5$ or average five independently seeded heads.

Granularity and direction of the self-calibrated null (Table S6). The self-calibrated null has two independent design choices, and the flag count depends on both. The *granularity* is whether an edge is corrected by its own replicate-measured ratio or by the ratio pooled over its whole system; per edge was pre-specified. The *direction* is whether the correction is applied only where the measurement says the bar is too tight, which can only remove flags, or in whichever direction the measurement points, which can also add them; the one-sided direction is the one the circularity concern of Section 4 calls for, and is the direction of Table S7. Table S6 reports all four combinations on the identical population, together with each arm’s realized probability of at least one false flag under a pure-null simulation in which every flag is a false positive by construction. The count of surviving flags ranges from two to sixteen across the four, so no single self-calibrated flag list is a stable object, and none of the four arms is reported as a flag set. Only the pre-specified per-edge one-sided arm holds its level: the two-sided arms fire far above nominal because rescaling an edge by its own replicate spread multiplies the statistic by σ^2/s^2 , whose expectation $\nu_{\text{rep}}/(\nu_{\text{rep}} - 2)$ is undefined at the $\nu_{\text{rep}} = 2$ replicate degrees of freedom three replicates provide. Taken together, the two-sided arms bound how far a flag list can move under calibration error of the size actually present, and the one-sided per-edge arm gives the conservative count: two of the six survive being retested against their own measured width. The realized levels are measured over 400 draws of the pure-null simulation specified above, so a realized 0.003 is a single flag and is not resolved from zero.

Table S6: **Self-calibrated cycle-closure flags by correction granularity and direction.** All four arms use the identical population (replicate 0, the 48 systems with at least one independent cycle) and the identical Benjamini–Hochberg level $\alpha = 0.05$; they differ only in whether the replicate-measured calibration ratio is applied per edge or pooled per system, and in whether it is applied only where it loosens a bar (one-sided) or in whichever direction the measurement points (two-sided). “published six” counts how many of the flags of Figure 4 the arm retains; “entering” lists the systems it adds. The last column is the realized probability of at least one false flag over 400 draws of a perfectly-calibrated synthetic null, against the nominal 0.05, with its exact binomial interval.

granularity	direction	flagged	published six	entering	null $P(\geq 1)$
reported se, uncorrected		6	all six	none	0.040 [0.023, 0.064]
per edge	one-sided	2	bace, cdk8	none	0.003 [0.000, 0.014]
per edge	two-sided	7	4 of 6	irak4_s2, thrombin, tyk2	0.588 [0.538, 0.636]
per system	one-sided	6	all six	none	not measured
per system	two-sided	16	all six	ciordia_retro, cmet, galectin, irak4_s2, keranen_p2, mcl1, renin, thrombin, tnks2, tyk2	0.240 [0.199, 0.285]

The per-edge two-sided arm retains **bace, brd4, cdk8** and **p38** and drops **faah** and **hif2a**. The per-edge one-sided arm loses more, retaining **bace** and **cdk8** alone; the two per-system arms retain all six. The per-system one-sided arm is the pre-registered circularity check of Table S7, whose harness reports the flag set rather than a synthetic-null level, so no realized probability is available for it; a one-sided correction can only widen bars, and the flag sets of both one-sided arms here are contained in the uncorrected one.

QC sweep under the real learned- σ profile (Figure S11). Figure 4B’s uniform $\times 0.15$ shrink is a stress test; here the learned head’s measured, overlap-dependent miscalibration profile (Figure 2A: ratio 0.09–0.20 $\times$ across overlap 0.26–0.78) is propagated through the identical GLS + Benjamini–Hochberg pipeline, per edge, by percentile-rank transfer onto each of 1143 (of the 1145; the sole cycle-free system contributes the other two) real edges’ own OpenFE **smallest_overlap** (an approximation: the two overlap statistics are on different scales, so only the profile’s ordering and spread are transferred, not its raw values). The rank transfer approximates how the head’s measured overconfidence would vary across these edges; it is not the head’s own output on this benchmark. Four arms are run on the same 48-system, 1143-edge network. The calibrated sandwich ($\times 1$) flags 6/48 (12%); the real profile (primary) flags 42/48 (88%), 7.00 $\times$ the calibrated count, past the pre-registered 2 $\times$ threshold; a shuffled-profile control (the identical per-edge ratios with their association to overlap destroyed, seed 20260808) also flags 42/48; and the uniform $\times 0.15$ stress test again flags 42/48. The three shrinking arms are not identical sets of systems (real vs. shuffled and real vs. uniform each swap exactly one system, **bace1** for **egfr**), so the per-edge ratio assignment is heterogeneous. The shuffled-profile control, the identical multiset of per-edge ratios with the association to overlap destroyed, also flags 42 of 48: the profile’s entire measured range (0.09–0.20 $\times$) is severely overconfident, every ratio is at most 0.20, so even the mildest value inflates an edge’s χ^2 contribution by at least $\sim 25\times$, and which systems cross the Benjamini–Hochberg threshold is set by the magnitude of the shrink rather than by its structure. The whole assignment is deterministic: each edge’s transferred ratio is a fixed function of its own overlap percentile, the shuffled control draws its permutation from the recorded seed, and the fit at each arm is a closed-form recomputation, so re-running reproduces the four arm counts exactly.

Dose-response of the QC to σ miscalibration (Figure S12). Figure 4B and Supplementary Figure S11 each characterize the cycle-closure test at a small number of scale points ($\times 1$ calibrated, $\times 0.15$ stress test / real profile); here a pre-registered, frozen grid of 16 uniform scale factors ($2.0\text{--}0.10\times$) is swept through the identical GLS + Benjamini–Hochberg pipeline on the same 48-system OpenFE network. Two readouts were frozen in advance: s_{onset} , the largest scale that flags strictly more systems than calibrated, and s_{50} , the largest scale at which at least half the systems are flagged. $s_{\text{onset}} = 0.9\times$ ($7/48$ vs. the calibrated $6/48$) is exactly one system crossing the Benjamini–Hochberg threshold. That readout is grid-determined rather than a measurement of this benchmark: because the flagged count is monotone in the scale, a strictly-exceeds-calibrated onset returns whichever grid point below 1.0 sits just past the first system’s threshold crossing. Independent probing places that system’s crossing as high as $s = 0.995\times$, so a finer grid would return $s_{\text{onset}} = 0.995\times$, and in the continuum limit $s_{\text{onset}} \rightarrow 1.0\times$ for any benchmark with a system near the cutoff. It is reported as the pre-registered readout. The informative readout is magnitude-based, such as the largest scale that doubles the calibrated count, which here is $0.7\times$ at $12/48$. The substantive change is mid-curve: from $0.7\times$ to $0.3\times$ the flagged fraction climbs from $12/48$ (25%) to $30/48$ (62%), reaching half the benchmark at $s_{50} = 0.3\times$ (the true 50% crossing is bracketed by the adjacent $0.35\times$ and $0.30\times$ grid points, which flag $23/48$ and $30/48$ respectively). The learned head’s measured, rank-transferred miscalibration band ($0.09\text{--}0.20\times$; Figure 2A, Supplementary Figure S11) lies entirely below s_{50} , so the head is not near the transition but well inside the saturated regime. The over-wide half of the grid is not flat either: relative to the calibrated $6/48$, a bar only 30% too wide ($1.3\times$) already loses five of the six flagged systems ($1/48$), and any wider bar ($1.5\times$, $2.0\times$) flags none, so the requirement is calibration, not conservatism. Two consistency checks confirm that the sweep reproduces the published test path: the $\times 1.0$ row reads $6/48$, matching Figure 4A’s calibrated count, and the $\times 0.15$ row reads $42/48$, matching Figure 4B and Supplementary Figure S11’s uniform/real-profile arms. The response to the shrink magnitude is graded rather than threshold-like. The quality-control test is therefore selective over roughly a factor of three of σ under-estimation below the calibrated scale, and over-wide σ costs power in the other direction, so the calibrated scale is a two-sided optimum rather than a plateau. The sweep is deterministic: the sixteen scale factors are a frozen grid and the test at each of them is a closed-form recomputation on fixed inputs, so re-running reproduces the entire curve exactly.

Per-system flag robustness to each system’s own calibration error (Table S7). The Figure 4 flag set raises a mild circularity concern: the per-system calibration of the sandwich null, measured independently against across-replicate spread (Figure 3), is heterogeneous, and four of the six flagged systems are themselves locally overconfident (**brd4** $0.76\times$, **faah** $0.90\times$, **bace** $0.96\times$, **hif2a** $0.96\times$; only **cdk8** $1.41\times$ and **p38** $1.43\times$ are conservative), so a flag could partly reflect that system’s own too-tight bars rather than genuine physical inconsistency. That concern is tested directly. For replicate 0, over the 48 systems with ≥ 1 independent cycle (the identical population as Figure 4), each system’s per-edge se is inflated by $1/\text{ratio}$ where its own ratio is below 1 (systems with ratio ≥ 1 are already conservative and are left unchanged), and the Benjamini–Hochberg test is recomputed over the whole set under this self-calibrated null. The criterion was fixed in advance: the result stands if every nominally-flagged system remains flagged, and is partial or negative if only some or none do. The population, the inflation rule and $\alpha = 0.05$ were fixed before this run. All six flags remain under their own self-calibrated null (Table S7), each self-calibrated reduced χ^2 ($2.36\text{--}8.97$) remaining far above the level a sampling-consistent network of this size and degree-of-freedom profile shows (Section 4, main text). This check pools the ratio over a whole system; the same one-sided inflation applied at the per-edge granularity this work pre-specified retains two of the

Table S7: **Per-system reduced χ^2 before and after inflating each system’s own per-edge standard error by its own measured calibration error.** “ratio” is the per-system root-mean-square ratio of reported standard error to across-replicate standard deviation of binding $\Delta\Delta G$, at $n = 3$. χ^2_{nom} is the nominal reduced χ^2 ; χ^2_{self} inflates that system’s own standard error by $1/\text{ratio}$ where $\text{ratio} < 1$ and leaves it unchanged otherwise. Rows are the six flagged systems.

System	ratio	χ^2_{nom}	χ^2_{self}	Still flagged
brd4	0.76	15.40	8.97	yes
bace	0.96	5.57	5.13	yes
faah	0.90	3.71	2.99	yes
cdk8	1.41	2.68	2.68	yes (unchanged; already conservative)
hif2a	0.96	2.53	2.36	yes
p38	1.43	2.40	2.40	yes (unchanged; already conservative)

six, and Table S6 reports both granularities in both directions. The distribution-free cross-replicate reproduction check of Figure 5B is a function of raw residual correlation only, never of se, so it does not depend on this null and is not affected by this concern in either direction. The per-system ratio in Table S7 is the root-mean-square reported standard error over the root-mean-square across-independent-replicate SD, pooled per system, not a naive per-system average of per-edge ratios or a naive ratio of arithmetic means, which for three of the six systems below would sit on the opposite side of $\text{ratio} = 1$. The ratios are raw $n=3$ SD ratios. Their residual small-sample bias is upward and target-specific, of order the reciprocal of the edge count, so debiasing would move all four of the locally overconfident flagged systems (**brd4** $0.76\times$, **faah** $0.90\times$, **bace** and **hif2a** $0.96\times$) further below one rather than toward it; the raw ratios used here are generous to those systems rather than the reverse. The computation is deterministic, and the full 48-system table is released with the analysis code.

Reward re-ranking over a measured congeneric set (Figure S13). Driving a trajectory-balance GFlowNet over **eg5** (the most σ -varying evaluable target), the calibrated lower-confidence-bound (LCB) reward lowers the true hit-rate (cal–raw -0.109), because on this set σ tracks chemical novelty ($\text{corr}(\sigma_{\text{total}}, \text{affinity}) = +0.42$) rather than low quality. An LCB reward improves the hit-rate only where σ flags low-quality candidates (Table S9 below).

The stopping decision as a function of the assumed error (Table S8). Figure S16 reports one point of a curve. The experiment measures how the stopping decision depends on the error the learner assumes, everything else held fixed, and the whole dependence is tabulated here. The $0.15\times$ arm of Figure S16 is a counterfactual stand-in: it imposes on the assumed error the miscalibration measured for a trained per-edge head in Figure 2, and is not an estimator run here. The two arms of the figure keep their own Monte-Carlo stream, so adding the sweep points leaves them unchanged.

The $0.94\times$ row is the 6% under-estimate a same-budget estimator pooling single-label edges by overlap reaches on the identical labels the heads were given (Figure 2). At that scale the stopping decision is indistinguishable from the calibrated one, 28.4 calls against 29.6 at the same 0.82 correctness. Correctness at the stop first moves at $0.60\times$, and the halved budget of Figure S16B needs $0.20\times$. The contrast is therefore scoped, exactly as the quality-control collapse of the main text is scoped, to per-edge heads missing by 5–11 $\times$, and is not evidence that a same-budget learned estimator would stop early. The one effect that survives a fair comparison is in the opposite direction: a 6% over-estimate spends 30.8 calls for the same correctness, the decision cost of a conservative

Table S8: **Simulated stopping under a swept assumed error.** Assumed posterior standard error as a multiple of the true one, against oracle calls at the stopping rule, top- k correctness at that stop, and the mean absolute gap between claimed confidence and actual correctness over the budget; 60 seeds. Starred rows are the two arms of Figure S16.

assumed se / true se	calls at stop	correct at stop	mean claimed – actual
1.06 $\times$ (conservative)	30.8	0.83	0.027
1.00 $\times$ (calibrated)*	29.6	0.82	0.022
0.94 $\times$	28.4	0.82	0.020
0.80 $\times$	25.4	0.82	0.029
0.60 $\times$	21.1	0.77	0.059
0.40 $\times$	17.9	0.68	0.093
0.20 $\times$	14.9	0.60	0.127
0.15 $\times$ (stand-in)*	14.4	0.60	0.137

bar of Figure S15B.

The decision cost of a conservative bar (Figure S15B). Even a conservative bar carries a decision cost: a systematic $\sim 1.41\times$ se inflation is itself a (safe-direction) miscalibration that, via the LCB rule, biases toward late stopping and under-commit. It errs in the safe direction against sampling noise, and because the independent-replicate SD is itself a lower bound on full reproducibility (Section 3), the inflation may be closer to neutral against the true run-to-run spread once pose and system-preparation variance are included.

Downstream audit summary (Table S9). Table S9 collects, for four downstream performance tasks (interval sharpness, active-learning edge weighting, commit-to-synthesis ranking, and reward re-ranking on the **eg5** congeneric set) and two decision tasks (commit trust and active-learning stopping), the calibrated method’s effect relative to the relevant baseline. The top block shows no measurable performance gain (Figures S14, S2, S15A, S13 above). The bottom block shows what the two decision tasks are measured against: an overconfident σ and a counterfactual stand-in are both far worse, which measures the cost of miscalibration, while against a fair baseline, the same overconfident σ after temperature or split-conformal recalibration and a same-budget pooled estimator at its measured 0.94 $\times$, the effect is absent (Figures S15B, S16 and Table S8 above).

Provenance of the experimental affinities. The grounding against measured affinity uses one ChEMBL target per system, verified to be a human single-protein target before use, and one assay type per system, taken in the order IC₅₀ then K_i and never mixed within a system. A flagged system enters the guided-versus-random comparison only if at least 60% of its network ligands map to a measured value. The counts here are grounded ligands, which for two systems exceed the ligand count of the sub-network the measurement runs on by one: an edge enters only when both its endpoints carry an affinity, so a ligand all of whose partners are ungrounded drops out. That is **1_1W7H** on **p38** (35 grounded, 34 in the sub-network) and **CAT-4j** on **bace** (24 and 23); **cdk8** and **hif2a** lose none. The **eg5** grounding of Section 4 predates that coverage-gated join and was assembled differently, per ligand from one named ChEMBL assay rather than by a target-wide structure search, so its row records the assay as well as the target. Table S10 lists what was used.

Table S9: **Effect of the decomposed σ relative to a baseline, across six downstream tasks.** Effects are the calibrated method minus the relevant baseline with a 95% bootstrap confidence interval; “real” denotes OpenFF benchmark out-of-distribution edges, “sim” denotes a controlled simulation. The two decision tasks are each given against an uncalibrated foil and against a fair baseline: the same σ recalibrated, and the assumed error a same-budget pooled estimator reaches.

Downstream task		Data	Effect (calibrated – baseline) [95% CI]			Verdict	
Interval sharpness (out-of-distribution)		real	width split/decomposed cond.	0.92 $\times$; -0.044 $[-0.121, 0.005]$		no gain	
Edge weighting (active learning)		sim	calls-to-top- k	$0.98\times$ $[0.93, 1.03]$		no gain	
Commit ranking (matched volume)		real	precision	-0.006 $[-0.016, +0.007]$		tie	
Reward re-ranking (evaluatable set, eg5)		real	true hit-rate	-0.109 $[-0.150, -0.072]$		hurts	(directional, single target) ^a
Commit trust vs overconfident σ		real	shortfall	$+0.182$ $[0.132, 0.225]$ better		cost of	miscalibration
Commit trust vs temperature-scaled σ		real	shortfall	$+0.034$ $[-0.018, +0.081]$		tie	
Commit trust vs split-conformal σ		real	shortfall	-0.029 $[-0.077, +0.007]$		tie	
Stopping vs stand-in	$0.15\times$	sim	82% vs 60% correct at stop			cost of	miscalibration
Stopping vs pooled estimator ($0.94\times$)		sim	0.82 vs 0.82 correct, 29.6 vs 28.4 calls			tie	

^aThe reward re-ranking effect is measured on a single evaluatable target (**eg5**) and is directional, and it is not yet established as a general property of calibrated reward re-ranking across chemotypes.

Table S10: **ChEMBL provenance of the experimental affinities.** Target identifier, protein, assay type, and the number of network ligands with a measured value, for every system grounded against measured affinity: the four flagged systems that clear the 60% coverage requirement, and **eg5**, the clean system used for the comparison of accuracy against experiment. Affinities are converted at $RT \ln 10 = 1.364$ kcal/mol and aggregated per ligand by the median pChEMBL value.

system	ChEMBL target	protein	assay	ligands
cdk8	CHEMBL5719	cyclin-dependent kinase 8	IC ₅₀	24
hif2a	CHEMBL1744522	endothelial PAS domain protein 1	IC ₅₀	41
p38	CHEMBL260	mitogen-activated protein kinase 14	IC ₅₀	35
bace	CHEMBL4822	beta-secretase 1	IC ₅₀	24
eg5	CHEMBL4581	kinesin-like protein KIF11	IC ₅₀	28

The **eg5** affinities were taken from a single named ChEMBL ATPase IC₅₀ assay, identifier CHEMBL1101849, queried per ligand; the other four rows come from a target-wide connectivity-based structure search under the 60% coverage rule, for which no single assay identifier applies.

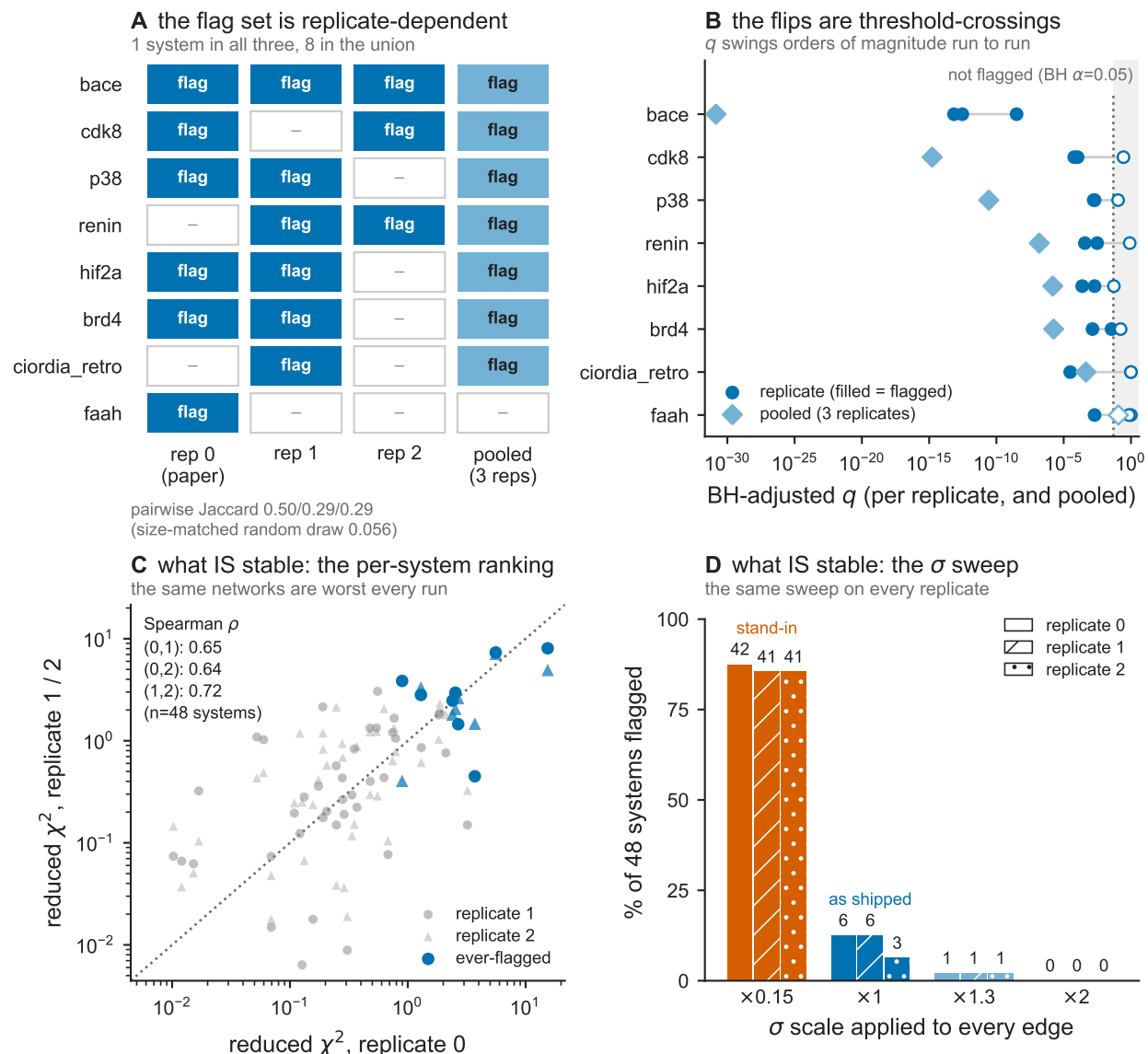


Figure S8: **Replicate stability of the flag set.** The Benjamini–Hochberg test at $\alpha = 0.05$ recomputed independently on each of the benchmark’s three protocol repeats. Per-system adjusted p -values, which systems are flagged on which replicate, the pairwise Jaccard overlap of the three flag sets against a random-draw reference, the flag set obtained from the three replicates pooled, and the selectivity of the test across a swept uniform σ scale on each replicate.

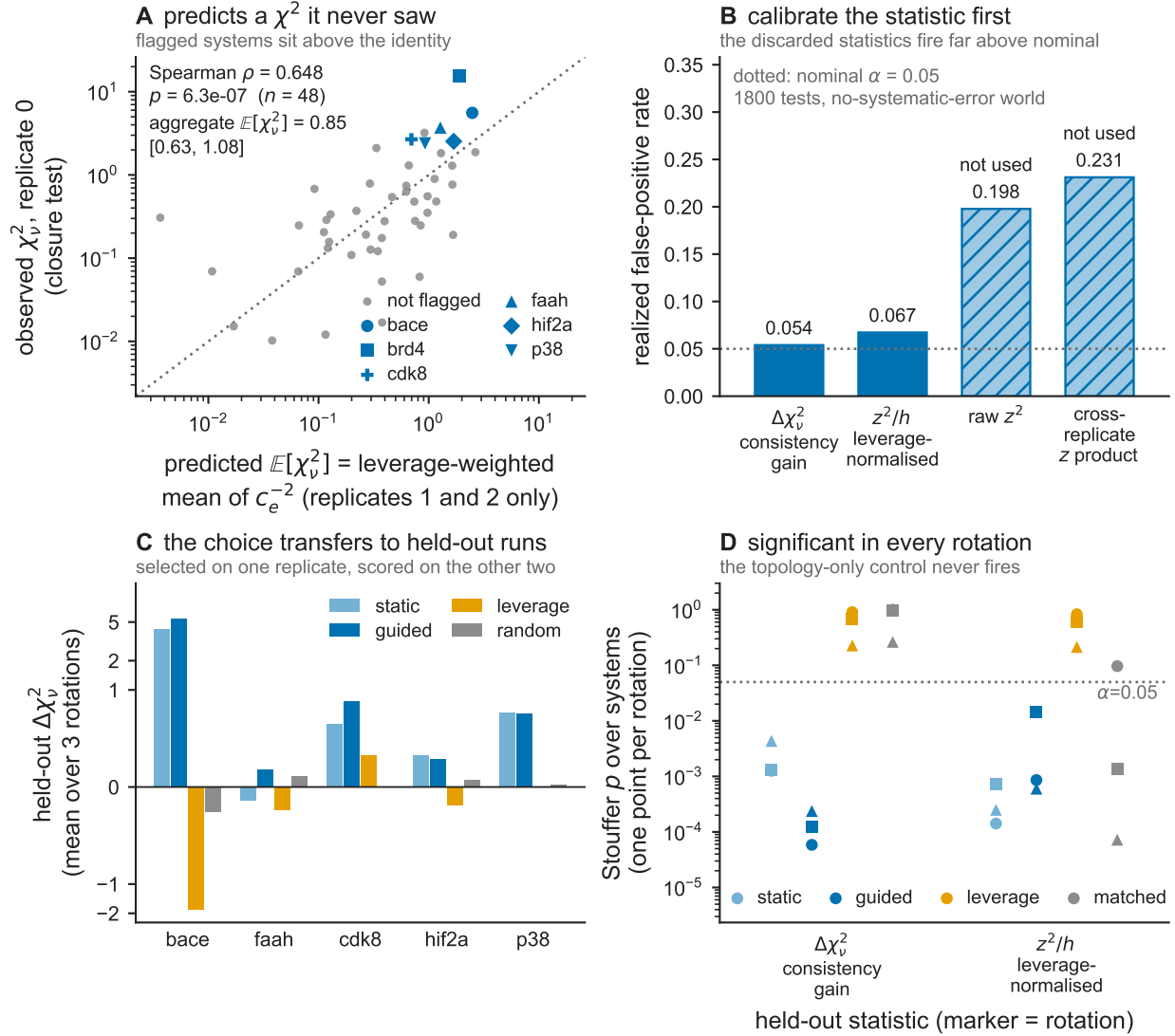


Figure S9: **Out-of-sample behaviour of the calibrated null.** (A) Per-system closure χ^2_v predicted from the per-edge miscalibration measured on the two replicates not being predicted, against the observed value, with the identity line and the flagged systems marked. (B) Realized false-positive rate of each candidate held-out statistic under a simulation with no systematic error anywhere, against the nominal level, with the two statistics that fire far above nominal marked. (C) Held-out change in reduced χ^2 on the two replicates not used for selection, after removing the edges selected on the third, averaged over the three rotations and shown per testable flagged system, for a static and a guided selection rule, a topology-only rule built from curl-leverage, and equal-size random removal. (D) Stouffer-combined p -value over systems, one point per rotation, for each of the two statistics that held their nominal level, under the same selection rules together with an $|z|$ -rank-matched control drawn from unflagged systems, with the $\alpha = 0.05$ line marked.

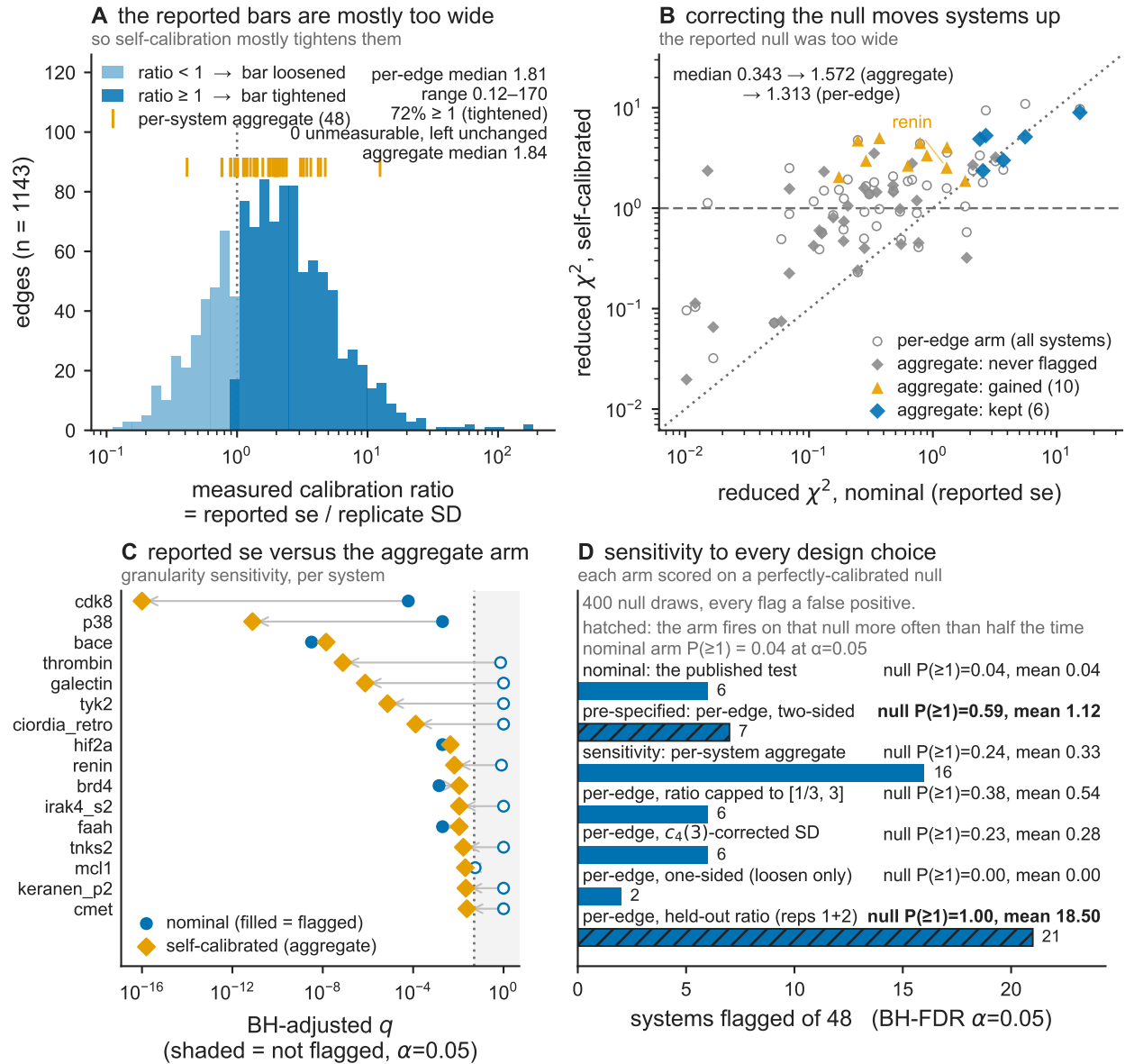


Figure S10: **Two-sided self-calibration of the per-edge error bar.** Per-system reduced χ^2 before and after replacing each system's reported standard error by its own replicate-measured calibration, which systems are gained and which are lost, and the realized probability of at least one false flag for each arm under a pure-null simulation, for the seven arms described in the text.

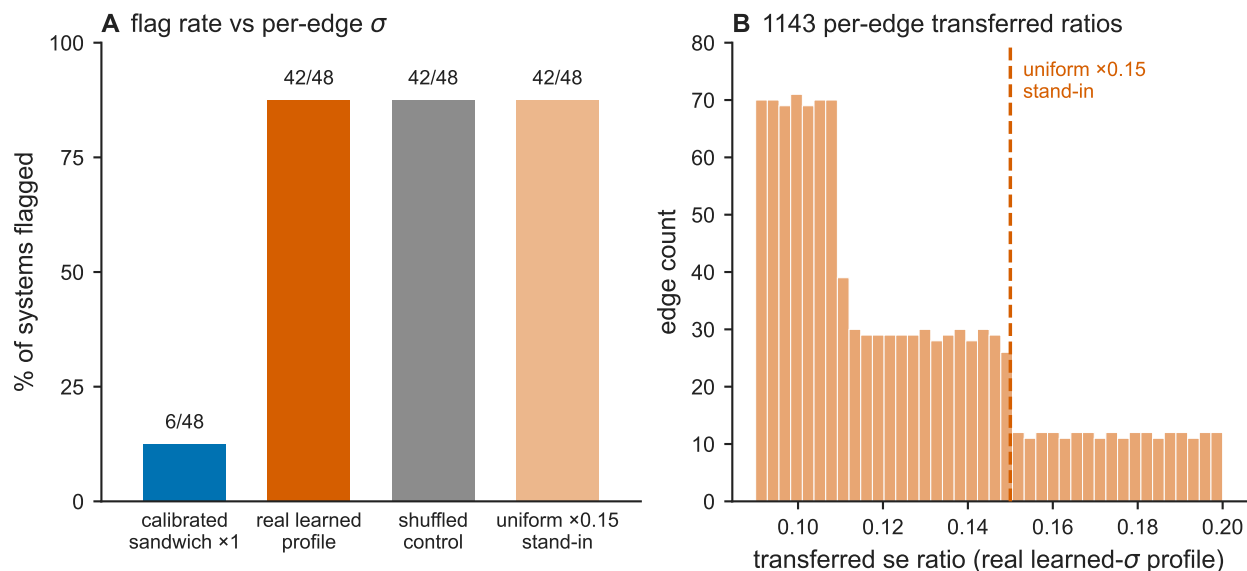


Figure S11: **Cycle-closure flagging under four per-edge σ assignments, and the distribution of the real-profile ratios.** (Left) Percentage of the 48-system OpenFE network flagged, for the calibrated sandwich, the learned head’s measured overlap-dependent miscalibration profile transferred by percentile rank onto each edge’s own overlap, a shuffled-profile control, and a uniform $\times 0.15$ shrink. (Right) Histogram of the 1143 per-edge transferred ratios from the real learned profile, with the uniform $\times 0.15$ stand-in marked.

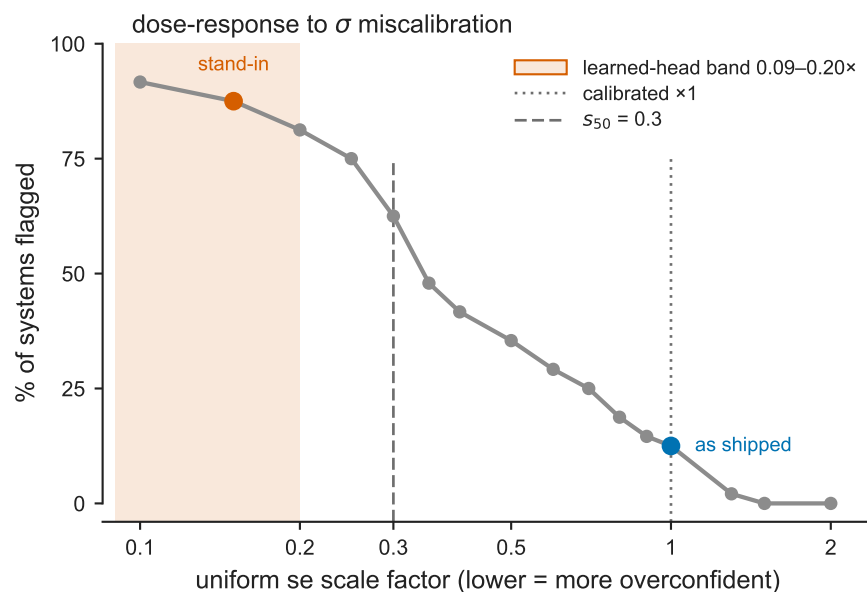


Figure S12: **Fraction of the OpenFE network flagged across a swept uniform σ scale factor.** Percentage of the 48-system network flagged by the cycle-closure test, across a frozen grid of 16 uniform standard-error scale factors from $2.0\times$ to $0.10\times$ (lower = more overconfident; $1.0\times$ is the calibrated sandwich). Vertical dotted line: the calibrated scale. Vertical dashed line: s_{50} , the largest scale at which at least half the systems are flagged. Shaded band: the learned head’s measured, rank-transferred miscalibration range.

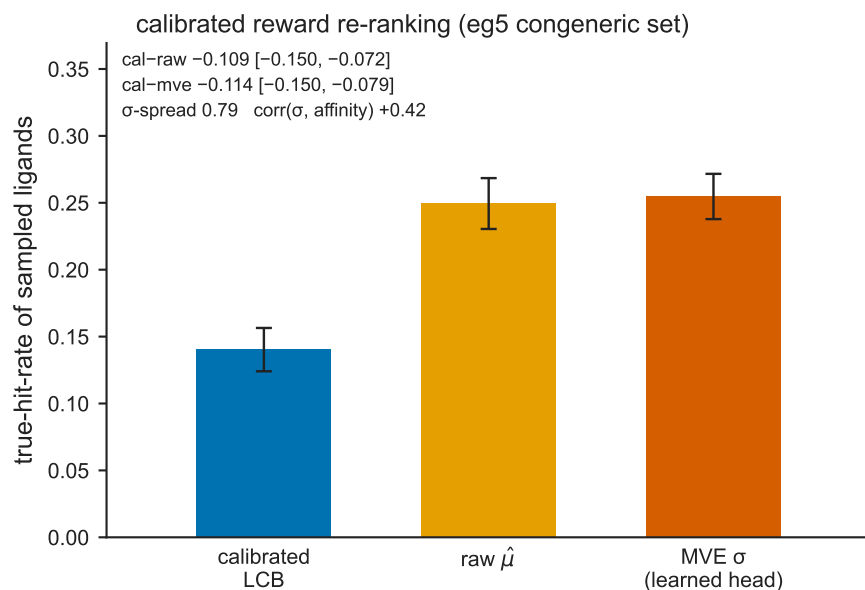


Figure S13: **True hit-rate of a trajectory-balance GFlowNet under three reward functions.** Mean true hit-rate of generated ligands, over 12 seeds with standard-error bars, for a calibrated lower-confidence-bound reward, a raw predicted-mean reward, and an overconfident MVE-uncertainty reward, on the **eg5** congeneric ligand series.

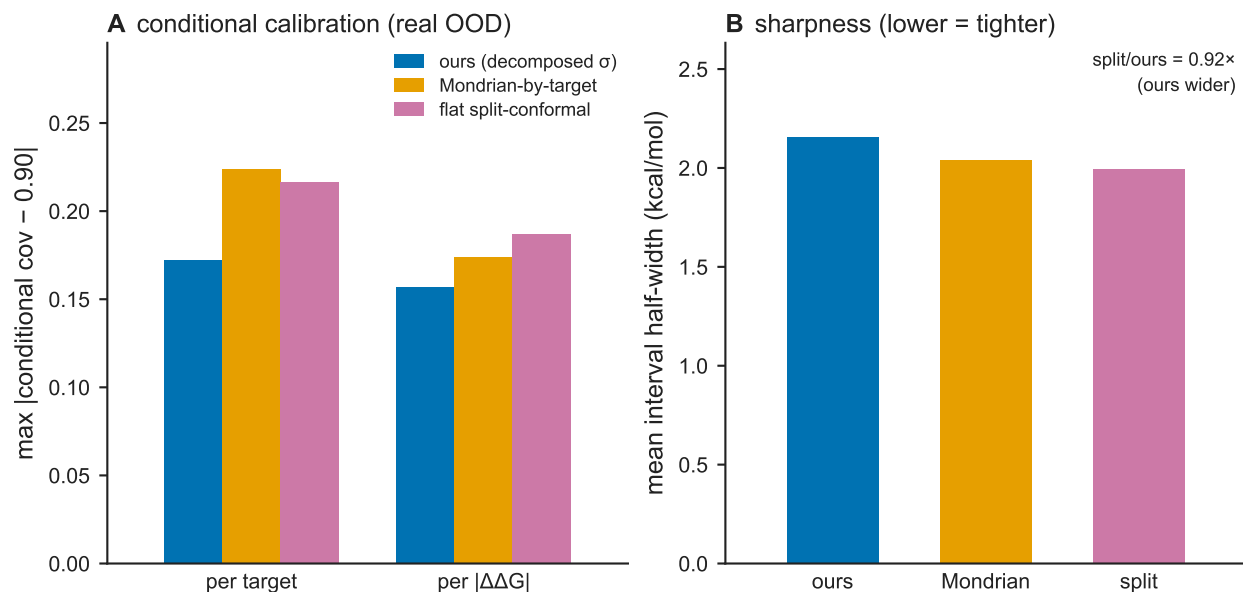


Figure S14: **Conditional calibration and interval sharpness on real out-of-distribution trunk residuals.** (A) Maximum conditional coverage error, computed per target and per $|\Delta\Delta G|$ bin, for the decomposed physics σ , a Mondrian-by-target conformal interval, and a flat split-conformal interval, each at approximately 90% marginal coverage. (B) Mean prediction-interval half-width for the same three methods.

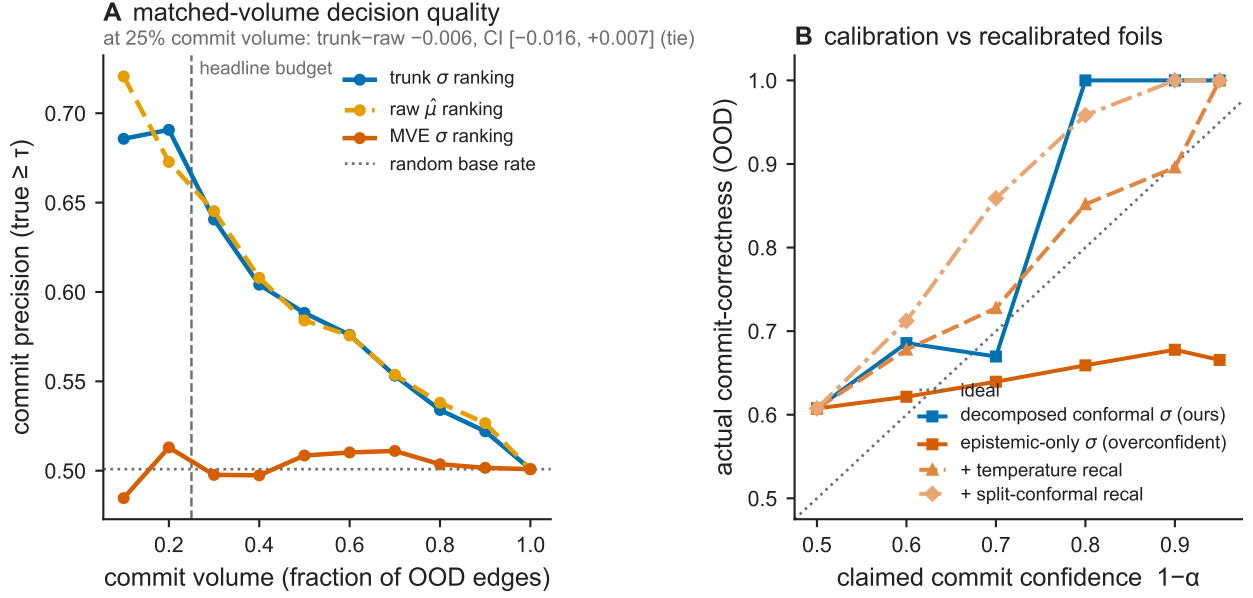


Figure S15: **Commit-to-synthesis ranking and calibration on real out-of-distribution edges.** (A) Commit precision (fraction of committed candidates with true value at or above a threshold τ) versus commit volume (fraction of edges committed), for candidates ranked by the trunk σ -aware margin, by the raw predicted mean, and by a learned MVE-head margin, with the random base rate marked. (B) Actual commit-correctness versus claimed commit confidence $1 - \alpha$, for the decomposed physics σ , an epistemic-only σ , and the same epistemic-only σ after temperature and after split-conformal recalibration, with the ideal (actual = claimed) line.

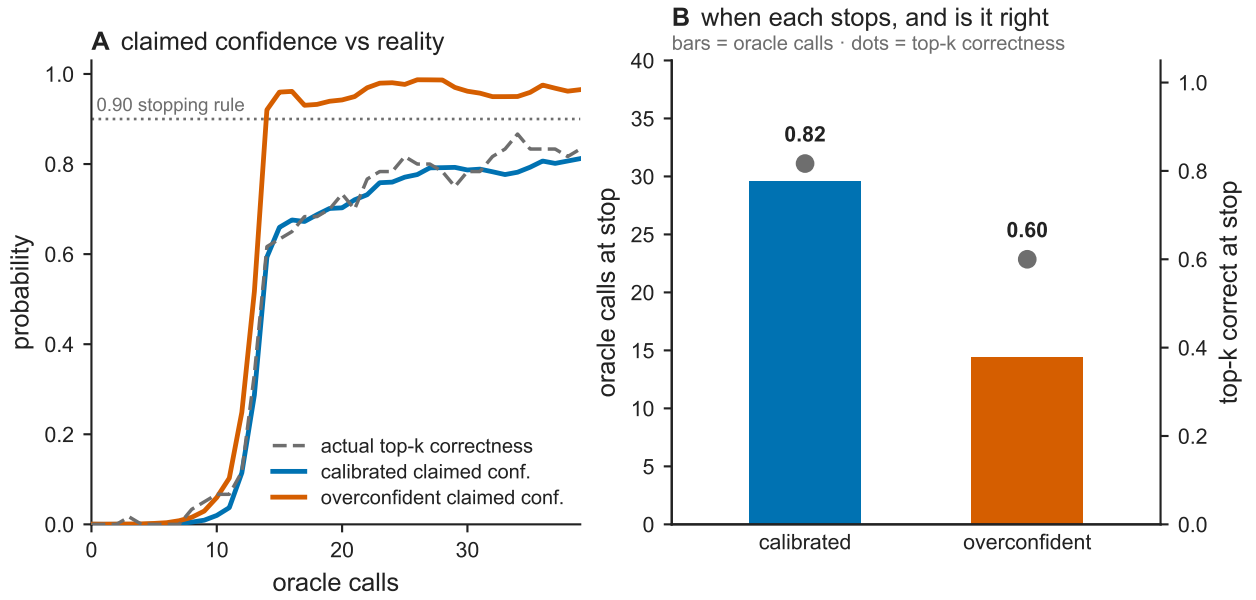


Figure S16: **Simulated active-learning stopping under calibrated versus counterfactually overconfident claimed uncertainty.** (A) Actual top- k correctness and claimed stopping confidence, averaged over 60 seeds, versus the number of oracle calls, for a calibrated learner and for a learner whose assumed posterior standard error is scaled by $0.15\times$. (B) Oracle calls at the stopping rule (bars) and top- k correctness at that stop (points), for the two learners.